



Daffodil
International
University

Course Description Handbook

Department of Multimedia & Creative Technology

Faculty of Science and Information Technology
Daffodil International University



Disclaimer

This Course Description Handbook has been issued in the absence of an officially published course handbook for the degree program. The document is based on approved curricula, course outlines, and academic information provided by the Department of Multimedia and Creative Technology, and accurately represents the structure, content, learning objectives, and academic scope of the program.

This handbook has been reviewed and formally approved by the concerned academic department. Its authenticity and institutional endorsement are certified through the signature of the Head of the Department.

Reviewed and approved by:



Md Salah Uddin

Assistant Professor and Head of the Department

Department of Multimedia & Creative Technology

Daffodil International University,

Daffodil Smart City (DSC), Birulia, Savar, Dhaka-1216

Email: headmct@daffodilvarsity.edu.bd, salah.mct@diu.edu.bd

Phone: +880 1847334769, +880 1515653128

Disclaimer	2
Department of Multimedia and Creative Technology (MCT)	6
Credit Requirement and Duration of the Program	7
Program Details	7
Course Description.....	11
MCT111: FUNDAMENTAL OF GRAPHIC DESIGN.....	11
MCT112: FUNDAMENTAL OF GRAPHIC DESIGN LAB.....	13
MCT113: DRAWING AND SKETCHING.....	15
MCT121: PRINCIPLES AND ELEMENTS OF DESIGN.....	17
MCT122: PRINCIPLES AND ELEMENTS OF DESIGN LAB.....	19
MCT123: COMPOSITION STUDY.....	21
MCT124: FIGURE DRAWING.....	23
MCT321: PHYSICS BASED ANIMATION.....	24
MCT133: BASIC TYPOGRAPHY DESIGN.....	26
MCT134: CALLIGRAPHY.....	28
MCT135: FILM STUDIES AND APPRECIATION.....	30
MCT211: 2D ANIMATION.....	31
MCT212: 2D ANIMATION LAB.....	33
MCT213: MOTION GRAPHICS.....	35
MCT221: 3D DESIGN AND MODELING.....	37
MCT224: BRAND DESIGN & PACKAGING.....	39
MCT233: VIDEO PRODUCTION.....	41
MCT311: AUDIO AND VIDEO STREAMING AND EDITING.....	43
CSE112: COMPUTER FUNDAMENTALS.....	44
AOL201: ART OF LIVING.....	46
MCT137: PERSPECTIVE STUDY.....	48
ENG113: BASIC FUNCTIONAL ENGLISH AND ENGLISH SPOKEN.....	50
ENG123: WRITING AND COMPREHENSION.....	52
MCT136: COMPUTER NETWORKS AND INTERNET.....	54
CIS112: MATHEMATICS-I FUNDAMENTALS OF MATHEMATICS.....	56
CSE122: STRUCTURED PROGRAMMING.....	58
CSE123: STRUCTURED PROGRAMMING LAB.....	60
MCT222: 3D DESIGN AND MODELING LAB.....	62
MCT223: MATHEMATICS-II COORDINATE GEOMETRY.....	64
GED201: BANGLADESH STUDIES.....	66
MCT231: OBJECT ORIENTED PROGRAMMING.....	68
MCT232: OBJECT ORIENTED PROGRAMMING LAB.....	70
CSE417: WEB ENGINEERING.....	72
CSE418: WEB ENGINEERING LAB.....	74
MCT313: 3D ANIMATION I.....	76
MCT314: CREATIVE WRITING.....	78
MCT315: 3D ENVIRONMENT DESIGN.....	80
CIS332: HUMAN COMPUTER INTERACTION.....	82

MCT321: COMPUTER AIDED DESIGN	84
MCT324: 3D ANIMATION 2	86
MIS575: PROJECT MANAGEMENT	87
MCT331: VISUAL EFFECTS PRODUCTION	90
MCT332: GRAPHIC NOVEL	92
MCT333: CHARACTER DESIGN & COMPOSITION (Z BRUSH).....	94
MCT334: SCRIPT PROGRAMMING	96
MCT335: CHARACTER ANIMATION LAB	97
MCT336: TEXTURING	99
CSE414: SIMULATION AND MODELING.....	102
CSE415: SIMULATION AND MODELING LAB.....	104
MCT413: NEW TYPEFACE DESIGN	106
MCT411: DIGITAL MATTE PAINTING.....	108
MCT421: GAME ANALYSIS AND DEVELOPMENT.....	110
MCT422: GAME ANALYSIS AND DEVELOPMENT LAB.....	112
MCT424: AUGMENTED AND VIRTUAL REALITY (AR/VR) DEVELOPMENT.....	114
MCT431: MULTIMEDIA AUTHORIZING.....	117
MCT432: DIGITAL CINEMATOGRAPHY.....	120
MCT433: LIGHTING AND RENDERING LAB.....	122
MCT450: LIGHTING AND RENDERING	124
MCT451: CHARACTER ANIMATION	126
MCT499: FINAL PROJECT.....	129

Department of Multimedia and Creative Technology (MCT)

The Department of Multimedia and Creative Technology is a championing department of Daffodil International University with an excellent group of highly creative students and faculty members. This is the only department in Bangladesh offering a bachelor's degree in Multimedia. The main objective of this department is to produce proactive multimedia professionals who have the potential to dominate the media industry at home and abroad.

From the period of its beginning, the Department of MCT has introduced those subjects and software in its course module which are acting as the shape-shifter in the multimedia industry throughout the world. This tendency enables us to create a man-force who are not only professionals but also specialists. Keeping the industry demand in mind DIU MCT is offering four major areas of specialization:

1. 3D Animation & VFX

Engineering

- a. 3D Modeling
- b. Environment Design
- c. Architectural Visualization
- d. Character Animation
- e. Stop Motion Animation
- f. Simulation & Modeling
- g. Visual Effects
- h. Digital Matte Painting

2. Visual Arts & Communication

- a. Drawing & Sketching
- b. Painting
- c. Graphic Design
- d. Typography
- e. Graphic Novel

- f. 2D Animation

- g. Motion Graphics

3. Game Design & Development

- a. Game Design
- b. Game Analysis
- c. Game Development
- d. Virtual Reality
- e. Web-Based

4. Film & Media

- a. Direction
- b. Filmmaking
- c. Script Writing
- d. Cinematography
- e. Pre-Production
- f. Post-Production

Credit Requirement and Duration of the Program

To obtain a B.Sc. in Multimedia and Creative Technology (MCT), students will have to complete 147 credits. The program has 12 semesters normally extending over four academic years.

Program Details

Students willing to obtain a bachelor's degree in Multimedia and Creative Technology (MCT) will have to follow the general guidelines of degree requirements of the university. The courses organized under four groups are shown in Table below:

Group	Group Name	Credits	Percentage
Group 1	Language and General Education	15	10.25%
Group 2	Basic Science	03	02.04%
Group 3	Mathematics	06	04.08%
Group 4	Core Courses	115	78.23%
Group 5	Elective Courses	08	05.44%

Group 1: Language and General Education (15 Credits)

Code	Course Title	Credit(s)	Prerequisite(s)
ENG113	Basic Functional English and English Spoken	3	NA
ENG123	Writing and Comprehension	3	ENG113
GED201	Bangladesh Studies	3	NA
MCT224	Brand Design and Packaging	3	MCT121
MIS575	Project Management	3	NA

Group 2: Basic Science (3 Credits)

Code	Course Title	Credit(s)	Prerequisite(s)
MCT321	Physics Based Animation	3	

Group 3: Mathematics (06 Credits)

Code	Course Title	Credit(s)	Prerequisite(s)
CIS112	Mathematics-I Fundamentals of Mathematics	3	NA
MCT223	Mathematics-II Coordinate Geometry	3	CIS112

Group 4: Core Courses (115 Credits)

Code	Course Title	Credit(s)	Prerequisite(s)
MCT111	Fundamentals of Graphic Design	3	NA
MCT112	Fundamental of Graphic Design Lab	1	NA
MCT113	Drawing and Sketching	3	NA
CSE112	Computer Fundamentals	3	NA
MCT121	Principles and Elements of Design	3	MCT111
MCT122	Principles and Elements of Design with lab	1	MCT112
MCT123	Composition Study	3	MCT113
AOL201	Art of Living	3	NA
MCT137	Perspective Study	1	MCT113
MCT211	2D Animation	3	MCT113
MCT136	Computer Networks and Internet	3	NA
MCT212	2D Animation lab	1	MCT121
MCT133	Basic Typography Design	3	MCT121
CSE122	Structured Programming	3	CSE112

CSE123	Structured Programming Lab	1	CSE112
MCT231	Object Oriented Programming	3	CSE122
MCT232	Object Oriented Programming Lab	1	CSE122
MCT213	Motion Graphics	3	MCT121
MCT124	Figure Drawing	1	MCT113
MCT221	3D Design and Modeling	3	NA
MCT222	3D Design and Modeling Lab	1	NA
MCT135	Film Studies & Appreciation	3	NA
MCT134	Calligraphy	1	MCT123
MCT313	3D Animation 1	3	MCT321
CSE417	Web Engineering	3	CSE231
CSE418	Web Engineering Lab	1	CSE232
MCT311	Audio and Video Streaming & Editing	3	NA
MCT332	Graphic Novel	3	MCT113
MCT324	3D Animation 2	1	MCT313
MCT314	Creative Writing	3	NA

CIS332	Human Computer Interaction	3	NA
MCT315	3D Environment Design	3	MCT321
MCT333	Character Design & Composition (Z brush)	3	MCT221
MCT334	Script Programming	1	CSE417
MCT336	Texturing	1	MCT221
MCT233	Video Production	3	MCT135
MCT331	Visual Effects Production	3	MCT233
MCT431	Multimedia Authoring	3	MCT334
CSE414	Simulation and Modeling	3	MCT451
CSE415	Simulation and Modeling Lab	1	MCT335
MCT413	New Typeface Design	3	MCT133
MCT432	Digital Cinematography	3	MCT233
MCT411	Digital Matte Painting	1	MCT233
MCT421	Game Analysis and Development	3	MCT431
MCT321	Computer Aided Design	3	NA
MCT422	Game Analysis and Development lab	1	MCT431

MCT424	Augmented and Virtual Reality Development (AR/VR Development)	3	MCT421, MCT422
MCT499	Final Project	6	

Group 5: Elective Courses (08 credits)

Code	Course Title	Credit(s)	Prerequisite(s)
MCT450	Lighting and Rendering	3	MCT321
MCT433	Lighting & Rendering Lab	1	MCT321
MCT451	Character Animation	3	MCT313
MCT335	Character Animation Lab	1	MCT313

Course Description

MCT111: FUNDAMENTAL OF GRAPHIC DESIGN

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

This course introduces the fundamentals of design, examining its definition, role, impact, and responsibility within society. It explores indigenous design practices and traces the evolution of visual communication from Gutenberg's movable type to contemporary digital media, including developments in printing and publishing technologies. Students build visual skills through sketching natural and man-made objects, construction and representational drawing, and basic design exercises. The course develops aesthetic awareness through the study of design elements such as line, shape, form, space, and color, along with principles including unity, balance, rhythm, perspective, emphasis, repetition, and proportion. Calligraphy, typography, and socially inspired motifs and patterns are introduced, alongside basic computer skills for creating, modifying, and presenting design work using digital tools, while fostering critical thinking, problem-solving, effective design communication, and positive work behavior.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply fundamental design knowledge effectively in practical and professional contexts.
- Develop solutions to real-world design problems through industry-oriented projects and assignments.
- Produce high-quality design work by incorporating feedback and refining outcomes to meet established standards.
- Use reading, research, and problem-solving skills to support informed design decisions.
- Collaborate effectively in team-based projects while demonstrating responsibility for individual contributions.

COURSE CONTENT

The course is structured to guide students from basic interface navigation to complex design execution including the following:

I) Introduction to Digital Tools

- Document setup including units, resolution, and color modes.
- Interface overview covering screen modes, zooming, panning, and basic layer management.
- Developing the ability to visualize creative ideas and translate them into digital formats.

II) Selection & Manipulation Techniques

- Mastery of selection tools including feathering and anti-aliasing.
- Application of advanced selection strategies for varied design scenarios.
- Understanding file formats and optimization for web and print output.

III) Retouching & Restoration

- Use of image retouching tools such as Spot Healing, Healing Brush, Patch Tool, Content-Aware Move, and Red Eye Tool for professional-quality results.

IV) Brush & Painting Engine

- Creation and customization of brushes and effective use of brush settings for artistic and corrective workflows.

V) Vector Integration

- Working with vector-based elements within raster software.
- Understanding the difference between bitmap and vector images.
- Utilizing vector drawing tools, paths, and tracing techniques for precise image construction.

VI) Project Development

- Execution of preliminary sketches and drafts into finalized digital designs.
- Application of design principles to socially and contextually relevant visual problems.
- Final project assessment encompassing all course topics through a comprehensive design project.

TEACHING & LEARNING METHODS

- Lecture Slide Presentations & Videos
- Brainstorming & Open Discussions
- Practical Workshops (Drawing & Digital Tools)
- Critique Sessions (Feedback on layouts)
- Gamification (Interactive content)

READING MATERIALS

- The Graphic Design Idea Book: Inspiration from 50 Masters by Steven Heller, Gail Anderson.
- Grid Systems in Graphic Design: A Visual Communication Manual by Josef Müller-Brockmann.

EVALUATION METHODS

- **Continuous Internal Evaluation (60%):** Includes Attendance, Class Tests, Assignments, Presentations, and Mid-Term Exams.
- **Semester End Examination (40%):** Consists of a final graphical project and theoretical assessment.

MCT112: FUNDAMENTAL OF GRAPHIC DESIGN LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

This course aims to introduce students to the practical foundations of graphic design through hands-on exploration of digital tools and visual communication systems. Emphasis is placed on understanding and applying principles of design such as unity, balance, rhythm, proportion, emphasis, and harmony, alongside calligraphy, typography, and socially inspired motifs and patterns. The course focuses on developing technical proficiency in image creation, manipulation, and presentation using industry-standard software, particularly Adobe Photoshop. Through guided lab exercises and projects, students learn to translate creative ideas into digital outcomes, preparing them with the essential skills required for entry-level positions in graphic design, multimedia, and related creative industries.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Navigate and utilize the Photoshop environment effectively for graphic design tasks.

- Apply design principles and digital tools to create, edit, and enhance images and compositions.
- Use selections, layers, masks, and retouching tools to produce professional-quality visual outputs.
- Prepare and optimize images for web, print, and presentation purposes.
- Develop complete design projects by integrating technical skills, visual aesthetics, and conceptual thinking.

COURSE CONTENT

- Introduction to the Photoshop environment, tools, panels, and workspace navigation
- Document setup including image size, resolution, and color modes
- Use of selection tools, masks, and refinement techniques
- Creation of photo collages and image composites
- Image enhancement and retouching using color correction and restoration tools
- Working with layers, blending modes, and layer masks
- Digital color application and adjustment techniques
- Use of filters and transformation tools for image manipulation
- Preparing and exporting images for web and print
- Understanding bitmap and vector graphics
- Image tracing and working with paths and vector tools
- Digital painting techniques and custom brush creation
- Execution of design projects from sketches to finalized digital outputs
- Final project development, presentation, and review

TEACHING & LEARNING METHODS

- Demonstration-based lab sessions
- Hands-on practical exercises
- Project-based learning
- Instructor-led critiques and reviews
- Peer discussions and guided feedback
- Online tutorials and reference-based learning
-

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Class participation and attendance
- In-class exercises and assignments
- Project work and presentations
- Mid-course reviews and final project submission

MCT113: DRAWING AND SKETCHING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

This course is designed to introduce students to the fundamentals of drawing with a focus on learning to observe, analyze, and represent forms accurately. It aims to develop perceptual and descriptive skills through exposure to a variety of drawing media, techniques, and subject matter while building a strong foundation in basic drawing methods and tools. The course encourages creative thinking, problem-solving, and critical analysis, guiding students to translate three-dimensional objects and spatial relationships onto two-dimensional surfaces. Students are supported in developing individual artistic expression, craftsmanship, and visual communication skills, while engaging collaboratively within a studio-based learning environment.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Demonstrate an understanding of the elements of art and principles of composition in drawing
- Apply contour lines, perspective, proportion, positive and negative space, and tonal values effectively
- Use a variety of drawing tools, techniques, and media with confidence
- Develop observational drawing skills through single-object and still-life studies
- Explore creative expression while maintaining aesthetic and professional standards

COURSE CONTENT

- Introduction to art, aesthetics, and drawing fundamentals
- Line practice, shapes, forms, and basic geometric construction
- Contour drawing and understanding positive and negative space
- Perspective, proportion, sighting, and measurement techniques
- Single-object drawing (bottles, vases) from observation
- Light and shade techniques to create depth and volume
- Still-life drawing using different media
- Sketchbook development and visual journaling
- Exploration of surface, texture, and material qualities

- Experimentation with traditional drawing media
- Development of individual style and creative expression
- Final still-life project integrating drawing fundamentals
- Presentation, critique, and review of completed work

TEACHING & LEARNING METHODS

- Studio demonstrations and guided practice
- Hands-on drawing exercises
- Group discussions and critique sessions
- Sketchbook-based exploration
- Visual references and observational studies

READING MATERIALS

Recommended Readings

- The Art of Basic Drawing by William F. Powell, Walter Foster Creative Team
- The Complete Guide to Asterix by Peter Kessler and Albert Uderzo

Supplementary Readings

- Bridgman's Complete Guide to Drawing from Life by George B. Bridgman

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and studio participation
- Assignments and sketchbook work
- Quizzes and mid-term examination
- Project reviews and presentations

Semester End Examination (40%), including:

- Final practical drawing examination
- Assessment of creativity, technique, and composition

MCT121: PRINCIPLES AND ELEMENTS OF DESIGN

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT111

TEACHING OBJECTIVES

This course introduces students to the foundational concepts of design, focusing on understanding what design is, how it functions, and how it is constructed using fundamental principles and elements. The course aims to stimulate design thinking and visual awareness, enabling students to analyze and interpret designs critically. Through historical perspectives, contemporary examples, and practical exploration, students develop the ability to break down designs into their core components and understand their effectiveness. Emphasis is placed on cultivating problem-solving skills, creative thinking, and industry-oriented design sensibilities, preparing students to create contemporary graphic designs that respond to real-world needs.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Understand and apply the principles and elements of design in visual communication
- Analyze and interpret designs critically using established design rules and frameworks
- Translate concepts and ideas into visual symbols and compositions
- Apply color psychology and visual hierarchy effectively in design solutions
- Create industry-relevant graphic design projects with creativity, clarity, and purpose

COURSE CONTENT

- Exploration of design from historical and contemporary perspectives to understand the evolution and everyday relevance of graphic design
- Understanding the distinction between graphic and graphics through discussion and examples
- Study of successful graphic designers' journeys and values derived from the design industry
- Introduction to the elements of design, including line, shape, form, space, texture, and color
- Understanding how design elements function individually and collectively to create effective compositions
- Hands-on demonstrations and experimentation with basic shapes and visual structures
- Study of color psychology and its impact on perception, emotion, and human behavior
- Development of confidence in applying color thoughtfully and purposefully in design
- Application of design principles through real-life, industry-oriented projects
- Integration of principles, elements, color theory, and conceptual thinking into visual solutions
- Final project assessing overall understanding, creativity, problem-solving ability, and professional presentation skills

TEACHING & LEARNING METHODS

- Lecture presentations and visual references
- Brainstorming sessions and open discussions
- Case studies and contemporary design analysis
- Practical demonstrations and guided exercises

- Project-based learning and critique sessions
- Interactive tools, gamification, and peer feedback

READING MATERIALS

Recommended Readings

- The Graphic Design Idea Book by Steven Heller and Gail Anderson

Supplementary Readings

- Grid Systems in Graphic Design by Josef Müller-Brockmann

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Final written and/or project-based assessment
- Evaluation based on conceptual clarity, analysis, application, and creativity

MCT122: PRINCIPLES AND ELEMENTS OF DESIGN LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT112

TEACHING OBJECTIVES

This lab course is designed as the practical extension of the Principles and Elements of Design theory course. Its objective is to enable students to apply theoretical knowledge of design principles and elements through hands-on digital practice. The course focuses on translating conceptual understanding into visual execution using vector-based design tools. Emphasis is placed on experimentation, problem-solving, and industry-oriented workflows, allowing students to develop practical competence in contemporary graphic design and strengthen their ability to analyze, construct, and present effective design solutions.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply principles and elements of design effectively in practical design projects

- Work confidently with vector and raster graphics in a digital design environment
- Use industry-standard software to create structured and visually balanced compositions
- Develop print-oriented design outputs such as brochures and desktop-published materials
- Execute real-world design projects following professional workflow and presentation standards

COURSE CONTENT

The lab begins with orientation sessions that introduce vector-based graphic software and reinforce digital workflows relevant to design practice. Students explore the integration of vector and raster graphics while gradually building proficiency in **Adobe Illustrator**, progressing from basic tool usage to advanced features. Through guided hands-on exercises and project-based learning, students apply design principles to print-oriented tasks such as brochure design and desktop publishing, gaining practical understanding of layout, typography, output settings, and production requirements. The course culminates in industry-inspired projects that challenge students to solve realistic design problems through iterative development, critique, and refinement, resulting in final projects that demonstrate the effective translation of conceptual design knowledge into professional visual outcomes.

TEACHING & LEARNING METHODS

- Demonstration-based lab instruction
- Hands-on practical exercises
- Project-based learning
- Guided critique and feedback sessions
- Industry-oriented design tasks

READING MATERIALS

Recommended Resources

- Adobe Illustrator Help and official documentation
- The Graphic Design Idea Book by Steven Heller and Gail Anderson

Supplementary Resources

- Grid Systems in Graphic Design by Josef Müller-Brockmann

Online Resources

- Adobe Illustrator Reference Guide
- Adobe Illustrator official website

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance and studio performance
- Practical exercises and lab work
- Project development and submission

- Final practical examination
-

MCT123: COMPOSITION STUDY

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT113

TEACHING OBJECTIVES

This course aims to develop students' understanding and application of composition as a foundational aspect of visual art and design, with a strong emphasis on visual literacy, perceptual awareness, and creative problem-solving. Students explore compositional principles through the study of line quality, mass, progression, form versus thought, space, proportional weight, and isometric scaling, along with perspective concepts such as foreground, middle ground, and background. The course engages students in the use of chiaroscuro, light and shade, surface studies, and medium control to create depth, emotional distance, and visual impact. Through experimentation with dimensional and mixed-media elements—including found objects, paper, and photographic extensions—students learn to build meaningful compositions informed by historical contexts, social issues, universal icons, and contemporary visual culture. The course fosters critical thinking by encouraging analysis, interpretation, and questioning of traditional art-making methodologies, while promoting collaboration, appreciation of diverse perspectives, and the development of creative intuition. Ultimately, students strengthen their visual thinking skills, artistic discipline, and problem-solving abilities, while gaining insight into potential career paths within art and related creative fields.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply principles of composition, drawing, and color theory in creating coherent visual works
- Demonstrate technical control and craftsmanship across various media
- Analyze and interpret compositions using appropriate critical and visual vocabulary
- Integrate visual thinking and creative problem-solving into finished artworks
- Engage in individual and collaborative artistic practices with awareness of potential art-related career paths

COURSE CONTENT

- Introduction to composition, art, and aesthetics

- Study of dots, lines, geometric and organic shapes, forms, and mass
- Visual balance, harmony, rhythm, progression, and proportional weight
- Perspective studies including foreground, middle ground, and background
- Light and shade (chiaroscuro) for depth, volume, and realism
- Realistic drawing of objects with varied materials, textures, and surfaces
- Still-life composition using monochrome and color approaches
- Exploration of creative composition through shape, color, and spatial arrangement
- Mixed-media and collage compositions using found objects and paper
- Close-up studies and watercolor-based compositional exercises
- Development of sketchbooks and visual journals for process documentation
- Application of composition to social themes, symbols, and visual narratives
- Final compositional project integrating concept, technique, and presentation
- Final project assessing overall understanding, creativity, problem-solving ability, and professional presentation skills

TEACHING & LEARNING METHODS

- Lecture presentations and visual demonstrations
- Studio-based drawing and composition exercises
- Sketchbook practice and guided experimentation
- Group discussions, critiques, and peer feedback
- Project-based learning and reflective analysis

READING MATERIALS

Recommended Readings

- The Art of Basic Drawing by William F. Powell, Walter Foster Creative Team
- The Complete Guide to Asterix by Peter Kessler and Albert Uderzo

Supplementary Readings

- Bridgman's Complete Guide to Drawing from Life by George B. Bridgman

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Assignments and sketchbook submissions
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Final practical and/or theoretical assessment based on compositional understanding, creativity, and execution

MCT124: FIGURE DRAWING

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT113

TEACHING OBJECTIVES

This course focuses on developing students' ability to observe, understand, and represent the human figure accurately and expressively. Emphasis is placed on gesture, proportion, anatomy, structure, and movement, enabling students to translate the living form onto a two-dimensional surface. Through theoretical understanding and sustained studio practice, students learn to analyze the human body using anatomical landmarks, perspective, light, and shadow. The course encourages disciplined observation, critical thinking, and professional studio habits while fostering confidence in drawing from live models, references, and imagination.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Draw the human figure accurately using proportion, perspective, and anatomical structure
- Apply gesture, line quality, and tonal values to convey movement and form
- Use light and shadow to describe volume and spatial depth
- Develop portrait and full-figure drawings from observation and imagination
- Demonstrate improved craftsmanship, discipline, and artistic confidence

COURSE CONTENT

- Introduction to figure drawing, art, and aesthetics
- Gesture drawing and quick studies to capture movement and posture
- Structural study of the human body and anatomical landmarks
- Detailed study of individual body parts and proportions
- Male and female figure anatomy and comparative analysis
- Block drawing and rapid figure sketching techniques
- Application of line quality, light, and shade in figure drawing
- Portrait drawing with emphasis on facial structure and expression
- Drawing from live models and reference material
- Use of various media including pencil, ink, brush, and color
- Development of sustained figure studies and final compositions
- Final figure drawing project integrating structure, anatomy, and expression

TEACHING & LEARNING METHODS

- Studio demonstrations and guided drawing sessions
- Observation-based practice with live models and references
- Short exercises and extended figure studies
- Critique sessions and instructor feedback
- Independent practice and project development

READING MATERIALS

Recommended Readings

- Bridgman's Complete Guide to Drawing from Life by George B. Bridgman
- The Art of Basic Drawing by William F. Powell, Walter Foster Creative Team

Supplementary Readings

- The Complete Guide to Asterix by Peter Kessler and Albert Uderzo

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance and studio discipline
- Lab performance and drawing exercises
- Lab reports and sketchbook submissions
- Final project and practical examination

MCT321: PHYSICS BASED ANIMATION

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT221

TEACHING OBJECTIVES

This course aims to develop a strong conceptual and practical understanding of the relationship between animation and the laws of physics. Students are introduced to fundamental principles of classical physics, particularly Newtonian mechanics, and their direct application in creating realistic three-dimensional animations. Emphasis is placed on understanding motion, force, gravity, momentum, and energy, and translating these concepts into believable animated behavior. The course also integrates the principles of animation with physics-based

simulations, enabling students to analyze real-world phenomena and recreate them accurately using contemporary 3D animation tools and simulation engines.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Upon successful completion of the course, students will be able to:
- Apply Newton's laws of motion to create physically accurate animations
- Animate objects demonstrating force, inertia, gravity, and momentum
- Create animations involving centripetal and centrifugal motion
- Integrate the principles of animation with physics-based motion
- Use dynamics and physics engines to simulate realistic object behavior

COURSE CONTENT

- Introduction to the three-dimensional world and its relationship to real-world physical systems, emphasizing the role of physics in animation
- Study and application of Newton's laws of motion, velocity, acceleration, circular motion, gravity, and conservation of momentum through basic animation exercises
- Exploration of different types of forces, including friction, air resistance, tension, work, and energy, to create force-driven animations
- Integration of the principles of animation—such as squash and stretch, anticipation, timing, arcs, follow-through, and appeal—with physics concepts to enhance realism and visual clarity
- Application of physics principles in character animation, including center of gravity and action–reaction relationships
- Hands-on practice with dynamics and simulation tools, including bouncing ball and rigid-body simulations
- Use of physics engines to create complete animations that accurately demonstrate physical laws through simulation-based workflows

TEACHING & LEARNING METHODS

- Lecture presentations and visual demonstrations
- Guided practical sessions in computer labs
- Physics-based animation exercises
- Workshops and project-based learning
- Critique sessions and iterative feedback

READING MATERIALS

Recommended Readings

- Physics for Animation by Arif Ahmed

Supplementary Readings

- Autodesk 3ds Max 2019 Complete Reference Guide by Kelly L. Murdock

EVALUATION METHODS

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Final project and/or written assessment demonstrating applied physics-based animation
-

MCT133: BASIC TYPOGRAPHY DESIGN

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT121

TEACHING OBJECTIVES

This course aims to develop an understanding of typography as both a cultural art form and a functional design discipline, with particular emphasis on the aesthetic, historical, and anatomical qualities of letterforms. It seeks to build sensitivity toward language, script, and typographic structure, especially in the context of Bengali typography. The course introduces students to essential typographic terminology, grammar, and principles of letter construction, enabling them to analyze, edit, and reinterpret typographic forms creatively. By engaging with historical, social, and technological influences on typography, students strengthen their foundational skills in font design, typographic composition, and layout, while preparing to apply typographic knowledge effectively within contemporary design and industry-oriented contexts.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Understand and use fundamental typography terminology and concepts
- Explain the historical evolution of typography with emphasis on Bangla type development
- Identify and analyze the anatomical structure of Bengali letterforms
- Apply principles of typography in editorial and layout-based design projects
- Design and refine basic font structures using systematic methods
- Demonstrate technical awareness of modern typographic systems and tools
- Develop creative, culturally informed typographic solutions suitable for professional use

COURSE CONTENT

- Introduction to language consciousness and the cultural significance of typography
- Historical overview of typography and the evolution of type systems
- Terminology of typography and foundational typographic principles
- Chronological development of Bangla typography and key contributors
- Transition from traditional to modern and digital typography practices
- Anatomical study of Bangla letterforms, including structure, proportions, and stroke logic
- Contemporary typography practices and technological influences
- Measurement, hierarchy, spacing, and complexity in typographic composition
- Newspaper and editorial design using typographic systems
- Masthead and custom typography design projects
- Detailed anatomical analysis of Bangla fonts and their historical transformations
- Font design methods, including flow drawing and construction logic
- Problem-solving in typography and typographic form correction
- Repetitive practice to master basic alphabet design and font consistency

TEACHING & LEARNING METHODS

- Lecture presentations and historical case studies
- Demonstrations and guided practice sessions
- Typography analysis and critique discussions
- Hands-on design exercises and project-based learning
- Seminars and interactive learning activities

READING MATERIALS

Recommended Readings

- The Complete Manual of Typography: A Guide to Setting Perfect Type 1st Edition by James Felici (Author), Frank Romano (Foreword)

Supplementary Readings

- Digital Calligraphy, Book by George Derwent Thomson
- Lettering and Modern Calligraphy: A Beginner's Guide: Learn Hand Lettering Complete Calligraphy by Hinkler Book.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Final written and/or design-based assessment focusing on typography concepts, analysis, and application
-

MCT134: CALLIGRAPHY

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT123

TEACHING OBJECTIVES

This course aims to introduce students to the art and practice of calligraphy as an expressive and disciplined form of visual communication closely connected to typography and design. The course develops an understanding of calligraphic principles, terminology, and structure, emphasizing precision, balance, and aesthetic control. Students learn to approach calligraphy as both a manual and digital practice, applying typographic sensitivity to create refined letterforms. The course encourages creative problem-solving, experimentation, and conceptual thinking, enabling students to produce expressive calligraphic works in both Bengali and English while maintaining strong design foundations.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Understand the fundamental concepts, terminology, and characteristics of calligraphy
- Apply calligraphic techniques to create single-word and multi-word compositions
- Design calligraphic headers, names, and pictorial type-based artworks
- Develop structured letterforms using character skeletons and proportional systems
- Execute calligraphic artworks manually and digitally with aesthetic and technical control

COURSE CONTENT

- Introduction to calligraphy through historical context and contemporary practice
- Study of calligraphic artists, professional workflows, and industry values
- Understanding the relationship between calligraphy, typography, and design
- Layout planning and guideline development for calligraphic compositions
- Character skeleton construction and scalable letterform development
- Practice of word-based and short-text calligraphy exercises
- Hands-on demonstrations of calligraphic tools, materials, and techniques
- Media selection and experimentation with ink, color, and surfaces
- Development of imagination and character reformation through conceptual layouts

- Final project demonstration, execution, and refinement
- Workshop-based experimental calligraphy projects
- Project submission, presentation, and critical review

TEACHING & LEARNING METHODS

- Lecture presentations and visual demonstrations
- Studio-based hands-on practice
- Workshops and guided exercises
- Critique sessions and peer feedback
- Project-based learning

READING MATERIALS

Recommended Readings

- Digital Calligraphy by George Derwent Thomson
- Lettering and Modern Calligraphy: A Beginner's Guide: Learn Hand Lettering

Supplementary Readings

- Complete Calligraphy by Hinkler Book.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance and studio participation
- Practical performance and exercises
- Lab reports and project documentation
- Final calligraphy project and presentation

MCT135: FILM STUDIES AND APPRECIATION

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

This course aims to develop students' ability to understand, analyze, and create visual narratives through the study of film language and production processes. It introduces the fundamentals of storytelling, scriptwriting, visual structure, and cinematic grammar, enabling

students to translate ideas into compelling screen-based narratives. The course emphasizes visual thinking, timing, shot division, and storyboard creation as essential tools for effective filmmaking. By engaging with the complete workflow—from story conception to editing and presentation—students gain a practical understanding of how films are planned, structured, and produced, bridging the gap between imagination, visualization, and final screen output.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Develop original stories and scripts using structured narrative techniques
- Convert written ideas into visual sequences through shot division
- Maintain effective timing and rhythm in film production
- Create storyboards and animatics as blueprints for production
- Demonstrate strong visual thinking and the ability to connect virtual ideas with visual outcomes

COURSE CONTENT

- Introduction to film studies, visual language, and the history of filmmaking
- Understanding the filmmaking process from concept to final production
- Story development from an idea, including story graphs and narrative structure
- Scriptwriting fundamentals and transformation of story into screenplay
- Principles of timing, pacing, and rhythm in visual storytelling
- Shot division techniques and conversion of text into visual scenes
- Storyboard creation as a visual blueprint for film production
- Animatic development for pre-visualization
- Editing fundamentals, including sequencing, timing control, and continuity
- Integration of sound, music, dialogue, and visual effects in production
- Full production editing workflow and post-production processes
- Trailer creation, transitions, and basic film marketing concepts
- Final presentation, review, and critique of film projects

TEACHING & LEARNING METHODS

- Lecture presentations and video screenings
- Visual demonstrations and case studies
- Practical exercises in storyboarding and editing
- Outdoor and studio-based sessions using camera and lighting
- Project-based learning and critique sessions

READING MATERIALS

Recommended Readings

- Film Studies: The Basics by Amy Villarejo

Supplementary Readings

- The Film Appreciation Book by Jim Piper

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Final project and/or written assessment demonstrating storytelling, visualization, and production understanding
-

MCT211: 2D ANIMATION

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT113

TEACHING OBJECTIVES

This course aims to develop students' skills in conceptualizing, designing, and producing two-dimensional animations using industry-standard software and workflows. Emphasis is placed on storytelling, visual aesthetics, and the principles of animation, guiding students from idea development and storyboarding to final animated output. The course introduces both traditional and cut-out animation techniques, enabling students to understand motion, timing, and energy in animation. By integrating creative thinking with technical execution, the course prepares students to apply 2D animation skills in areas such as motion graphics, digital storytelling, and portfolio development.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Identify and apply the twelve principles of animation through critical analysis of animated works
- Develop strong animated narratives through effective story development and storyboarding
- Calculate and apply appropriate frame rates and timing in animation production
- Create aesthetically appealing 2D animations demonstrating controlled motion and energy
- Use industry-standard animation tools to produce traditional, cut-out, and computer-generated animations

- Analyze historical and contemporary animation styles and trends

COURSE CONTENT

- Introduction to 2D animation, instructional methodology, and professional animation practices
- Historical overview of 2D animation and evolution of animation techniques
- Familiarization with 2D animation software and production pipelines
- Story development fundamentals, including setting, character, plot, conflict, and resolution
- Importance of narrative structure and storytelling in animation
- Study and application of the twelve principles of animation
- Story finalization and planning for animated projects
- Preparation of animation assets using vector-based software
- Integration of assets into animation software and symbol creation
- Layer management, libraries, and workflow optimization
- Frame-by-frame and cut-out animation techniques
- Experimental use of layers and motion techniques
- Group-based project development and collaborative production
- Review, problem-solving, publishing, and output settings
- Completion and presentation of a final 2D animation project

TEACHING & LEARNING METHODS

- Lecture presentations and video demonstrations
- Software-based practical sessions
- Storyboarding and animation exercises
- Group discussions, critiques, and feedback sessions
- Project-based learning and presentations

READING MATERIALS

Recommended Readings

- The Animator's Survival Kit by Richard Williams

Supplementary Readings

- Producing Independent 2D Character Animation: Making & Selling A Short Film
- Animation Writing and Development, From Script Development to Pitch (Focal Press)

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Final animation project and/or theoretical assessment based on concept, execution, and aesthetics

MCT212: 2D ANIMATION LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT121

TEACHING OBJECTIVES

This lab course is designed as the practical extension of the 2D Animation theory course, focusing on the full implementation of previously acquired conceptual and technical knowledge. The course aims to immerse students in a professional animation production environment, guiding them through the complete 2D animation pipeline from story selection to final output. Emphasis is placed on teamwork, production management, advanced animation techniques, and problem-solving within real production constraints. By working on group-based projects, students gain hands-on experience in traditional, cut-out, and tradigital animation workflows, preparing them for portfolio development and industry-oriented practice.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply the twelve principles of animation effectively in complete animation projects
- Produce accurate and aesthetically appealing 2D animations using industry-standard tools
- Manipulate animation production tools, rigs, and assets efficiently
- Manage multiple scenes, layers, and assets within a single production file
- Critically analyze, evaluate, and refine animated work based on feedback and industry standards
- Collaborate effectively in team-based animation production environments

COURSE CONTENT

- Continuation and refinement of projects initiated in the 2D Animation theory course
- Overview of the complete 2D animation production pipeline
- Team formation, role assignment, and production management practices
- Advanced storyboarding techniques, including shot division and animatic refinement
- Character skeleton construction and rigging using bone tools

- Frame-by-frame and cut-out animation techniques
- Handling complex scenes, symbols, and large-scale layer management
- Render settings, publishing formats, and output optimization
- Iterative project reviews, problem identification, and solution development
- Asset review including animatics, props, voice, and sound integration
- Final project submission, presentation, and documentation

TEACHING & LEARNING METHODS

- Studio-based hands-on animation sessions
- Guided demonstrations and workflow walkthroughs
- Group-based project production
- Review meetings, critiques, and feedback sessions
- Problem-solving workshops and production reviews

READING MATERIALS

Recommended Readings

- The Animator's Survival Kit by Richard Williams Expanded Edition.

Supplementary Readings

- Producing Independent 2D Character Animation: Making & Selling A Short Film
- Animation Writing and Development, From Script Development to Pitch (Focal Press)

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance and studio discipline
- Lab performance and practical execution
- Project documentation and lab reports
- Final group animation project and presentation

MCT213: MOTION GRAPHICS

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT121

TEACHING OBJECTIVES

This course aims to introduce students to the fundamentals and professional practices of motion graphics, focusing on animating static visual elements through time-based motion and design principles. Emphasis is placed on understanding animation principles, keyframe-based motion, and visual aesthetics using industry-standard software. The course develops technical proficiency in text animation, shape-based motion, compositing, and simple character animation, while also introducing 2D, 3D, and real-time footage integration. By combining creativity with technical control, students learn to design and produce complete motion graphic sequences suitable for advertising, digital media, and contemporary visual communication.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Navigate and use the Adobe After Effects interface efficiently
- Apply animation principles and keyframe techniques to create custom 2D animations
- Manage 2D and 3D layers for structured motion graphic compositions
- Create text animations, shape animations, and path-based motion graphics
- Perform basic 2D character animation using tools such as the Puppet tool
- Apply motion tracking, masking, color grading, and lighting techniques
- Composite animated elements with real-time footage
- Export and render motion graphic projects in appropriate audio and video formats

COURSE CONTENT

- Introduction to motion graphics and its applications in industry and media
- Principles of animation and their application in motion design
- User interface and workflow of Adobe After Effects
- Basic transform animations using scale, position, rotation, opacity, and anchor point
- Layer types and layer management, including solid, adjustment, null, and shape layers
- Text animation techniques and typographic motion design
- Masking, path animation, and stroke-based custom animations
- Advanced keyframe animation using the graph editor for smooth motion control
- Motion tracking and custom keyframe tracking
- Color grading and visual enhancement techniques
- Lighting effects, glow, and visual depth creation
- Importing and animating vector assets from Adobe Illustrator
- Simple 2D character animation using the Puppet tool
- Working with 3D layers, cameras, and composition space
- Understanding composition grammar and visual balance in motion design
- Blending modes, track mattes, and real-time footage compositing
- Exporting and rendering using After Effects and Media Encoder
- Final project development, presentation, and evaluation

TEACHING & LEARNING METHODS

- Lecture presentations and video demonstrations
- Software-based practical sessions in computer labs

- Guided animation exercises and demonstrations
- Project-based learning and critiques
- Interactive discussions and feedback sessions

READING MATERIALS

Recommended Readings

- Design for Motion: Fundamentals and Techniques of Motion Design 1st Edition by Austin Shaw (Author).

Supplementary Readings

- The Animator's Survival Kit: A Manual of Methods, Principles, and Formulas for Classical, Computer, Games, Stop Motion and Internet Animators (FARRAR, STRAUS) Paperback – Illustrated, September 25, 2012 by Richard Williams (Author).

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Final motion graphics project and/or theoretical assessment based on creativity, execution, and technical accuracy

MCT221: 3D DESIGN AND MODELING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

This course aims to introduce students to the fundamentals of three-dimensional design and modeling as the foundation of 3D animation, CGI, and visual effects production. Using industry-standard software Autodesk Maya, the course develops a clear understanding of 3D workflows, tools, and techniques required to create models, environments, textures, materials, and rendered outputs. Emphasis is placed on building strong modeling fundamentals, understanding form and topology, and applying technical skills to create production-ready 3D

assets. By combining conceptual design principles with hands-on software practice, the course prepares students for advanced animation, simulation, and visual effects courses.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Create 3D models using primitive, polygon, and NURBS modeling techniques
- Apply industry-standard modeling workflows and topology principles
- Use Autodesk Maya efficiently for modeling, texturing, lighting, and rendering
- Develop and apply materials and textures using UV mapping techniques
- Set up cameras, lighting, and scenes for still and animated renders
- Perform basic animation and shot setup within a 3D environment
- Render final outputs suitable for presentation and production use

COURSE CONTENT

- Introduction to 3D design, modeling pipelines, and asset production workflows
- Overview of different modeling approaches, including polygon, NURBS, subdivision, sculpting, procedural, modular, and simulation-based modeling
- Understanding Autodesk Maya interface, navigation, workspace customization, and viewport controls
- Working with primitives, transformations, duplication methods, grouping, pivots, and object hierarchy
- Polygon modeling fundamentals using vertices, edges, faces, extrude, bevel, bridge, and edge loops
- Boolean operations and modifier-based modeling techniques
- Topology principles, mesh optimization, and clean geometry practices
- Advanced modeling tools including soft selection, deformers, lattice, mirror, loft, and lathe techniques
- Introduction to NURBS modeling using curves and surfaces
- Intermediate modeling workflows including sculpting, retopology, quad draw, and mesh cleanup
- Importing and integrating external assets, including AI and SVG files
- Texturing fundamentals using UV mapping and material assignment
- Basic lighting, camera setup, rendering, and playblast previews
- Creation of complete 3D scenes and presentation-ready assets

TEACHING & LEARNING METHODS

- Lecture presentations and video demonstrations
- Hands-on practical sessions in computer labs
- Guided modeling exercises and assignments
- Project-based learning and critiques
- Industry workflow discussions and case studies

READING MATERIALS

Recommended Readings

- Autodesk 3ds Max 2019 Complete Reference Guide by Kelly L. Murdock

Supplementary Readings

- Physics for Animation by Arif Ahmed

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Final practical and/or theoretical assessment demonstrating modeling, texturing, and rendering skills
-

MCT224: BRAND DESIGN & PACKAGING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT121

TEACHING OBJECTIVES

This course aims to develop an in-depth understanding of branding as a philosophical, psychological, and visual communication system rather than merely a product identifier. The course introduces students to brand thinking, strategy, and positioning while translating abstract brand values into tangible visual identities and packaging solutions. Emphasis is placed on conceptual development, research, visual storytelling, and three-dimensional thinking in packaging design. Through studio projects and real-world case studies, students learn to design coherent brand systems, corporate identities, and packaging that effectively contain, present, and communicate products within contemporary industry and market contexts.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Explain the conceptual, psychological, and strategic foundations of branding
- Analyze and apply branding terminology, principles, and positioning strategies
- Design brand names, logos, and corporate identity systems
- Develop visual identities aligned with brand philosophy and target audiences

- Create functional and aesthetic packaging solutions, including FMCG packaging
- Produce a complete brand project integrating logo design, identity systems, and packaging

COURSE CONTENT

- Introduction to branding concepts, philosophy, and industry values
- Understanding brands beyond products through real-world examples
- Brand strategy, brand management, and contemporary branding practices
- Brand positioning, brand elements, and market differentiation
- Challenges in brand establishment and methods of resolution
- Research-driven project development and brand analysis
- Brand naming strategies and critical evaluation of brand names
- Logo design process, layout development, and refinement
- Typographic considerations in logo design, including Bangla font anatomy
- Finalization of logos and resolution of typographic challenges
- Development of corporate identity systems including color, typography, and visual elements
- Packaging design fundamentals with emphasis on structure, usability, and visual appeal
- Creation of packaging mockups and brand touchpoints
- Development of concise brand guidelines and final project presentation

TEACHING & LEARNING METHODS

- Lecture presentations and case study analysis
- Concept development and visual research
- Studio-based design exercises and critiques
- Project-based learning with iterative feedback
- Industry-oriented discussions and branding analysis

READING MATERIALS

Recommended Readings

- Packaging Design: Successful Product Branding from Concept to Shelf by Marianne Klimchuk & Sandra Krasovec

Supplementary Readings

- Brand Bible edited by Debbie Millman
- Branding the Nation by Melissa Aronczyk

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Project presentations and mid-term examination

Semester End Examination (40%), including:

- Final branding and packaging project and/or theoretical assessment
-

MCT233: VIDEO PRODUCTION

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT135

TEACHING OBJECTIVES

This course aims to introduce students to the complete process of digital video production, from conceptual development and pre-production planning to production and post-production execution. Emphasis is placed on visual storytelling, teamwork, technical proficiency, and aesthetic decision-making. Students learn to research and develop project ideas, plan shots and storyboards, operate cameras and lighting equipment, and edit video using non-linear editing systems. Through hands-on projects and collaborative workflows, the course enables students to explore different video genres, refine their creative and technical skills, and gain insight into professional practices and career pathways in video and media production.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Write, plan, shoot, direct, and edit video projects using professional workflows
- Apply pre-production, production, and post-production techniques effectively
- Demonstrate knowledge of video genres, shooting styles, and field production methods
- Operate cameras, lenses, lighting, and audio equipment with technical accuracy
- Edit video using non-linear editing software, including audio, graphics, and transitions
- Develop personal creative direction and identify potential career paths in video production

COURSE CONTENT

- Introduction to video production and professional industry practices
- Overview of video formats, genres, and shooting styles
- Pre-production fundamentals including research, scripting, storyboarding, and planning
- Production techniques covering camera operation, framing, movement, and sound capture
- Introduction to non-linear editing systems and project organization

- Video editing workflows including timeline editing, audio synchronization, transitions, and titles
- Text animation for video intros and outros
- Lighting techniques for studio and field production
- Understanding lenses, cameras, and essential accessories
- Color correction and color grading for visual consistency
- Audio and video compositing, narration, and music integration
- Exporting and delivering video for web, broadcast, and presentation
- Final project development, presentation, and critique

TEACHING & LEARNING METHODS

- Lecture presentations and video demonstrations
- Practical shooting sessions and outdoor production exercises
- Editing workshops using professional software
- Group-based projects and role-based teamwork
- Critique sessions and guided feedback

READING MATERIALS

Recommended Readings

- How to Shoot Video That Doesn't Suck: Advice to Make Any Amateur Look Like a Pro by Steve Stockman

Supplementary Readings

- Corporate Video Production: Beyond the Board Room by Stuart Sweetow

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Project presentations and mid-term examination

Semester End Examination (40%), including:

- Final video project and/or theoretical assessment demonstrating production competence

MCT311: AUDIO AND VIDEO STREAMING AND EDITING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40

Prerequisite(s) : None

TEACHING OBJECTIVES

This course aims to introduce students to the fundamental principles and professional practices of audio and sound design for visual media. Emphasis is placed on understanding sound as a critical component of visual communication and its role in film, animation, games, streaming media, and digital content. The course develops both theoretical and practical knowledge of sound, digital audio, music composition, and audio post-production. Students learn how to record, design, edit, mix, and master audio, as well as how to creatively synchronize sound with visuals. By combining technical training with aesthetic awareness, the course prepares students to create impactful soundtracks and audio solutions for diverse media platforms.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Identify, describe, and analyze sound and music in visual productions
- Understand the fundamentals of sound, digital audio, and music theory
- Create and manipulate audio using digital audio workstations (DAWs)
- Apply sound effects, music, and dialogue creatively and technically
- Design, edit, mix, and master audio for film, animation, games, and streaming media
- Develop critical listening skills and enhanced musical sensibility

COURSE CONTENT

- Introduction to sound, audio concepts, and object-oriented audio thinking
- Listening, analysis, and evaluation of sound in professional media examples
- Fundamentals of sound, frequency (Hz), waveforms, and digital audio concepts
- Introduction to DAWs and beat-making workflows
- Music computing fundamentals including psychoacoustics and sound perception
- VST instruments, sound balancing, and automation techniques
- Groove and melody creation using piano roll and virtual instruments
- Understanding music genres, audio formats, bandwidth, and frequency range
- Diegetic and non-diegetic sound in film and media
- Sound recording techniques including Foley and soundtrack elements
- Music composition, sound design, and automation workflows
- Audio editing, mixing, and multi-track sound balancing
- Advanced sound mixing using VSTs and mixing panels
- Final audio production, presentation, and critical review

TEACHING & LEARNING METHODS

- Lecture presentations and video demonstrations
- Listening sessions and analytical discussions
- Software-based practical sessions using DAWs
- Studio and workshop-based audio production exercises
- Project-based learning and critique sessions

READING MATERIALS

Recommended Readings

- Fundamentals of Musical Composition by Arnold Schoenberg
- A Practical Guide to Musical Composition by Alan Belkin

Supplementary Readings

- Harvard Dictionary of Music by Willi Apel
- Audio & Music Production Guide edited by Fahim Faisal

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests, quizzes, and assignments
- Project presentations and mid-term examination

Semester End Examination (40%), including:

- Final audio production project and/or theoretical assessment
-

CSE112: COMPUTER FUNDAMENTALS

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

Computer Fundamentals is a foundation course designed to prepare MCT students for advanced courses such as Structured Programming, Object-Oriented Programming, Scripting, Game Development, Animation, and Multimedia Applications. The course introduces students to the basic principles of computing, logical program flow, and problem-solving using conditions, loops, and flowcharts. It emphasizes how data and instructions are processed by computer systems, the role of memory and storage, and the importance of number systems and coding schemes. Students also gain essential knowledge of computer hardware, software, operating system concepts, computer networks, security fundamentals, and professional productivity tools such as Google Workspace. The course builds both conceptual understanding and practical skills required for effective use of modern computing systems.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Understand data, information, variables, and logical program flow
- Apply loops, conditional statements, and flowcharts to solve basic problems
- Identify and correct logical errors in simple programs and flowcharts
- Explain computer hardware components, memory hierarchy, and storage systems
- Understand number systems, computer codes, logic gates, and microcomputer concepts
- Describe input/output devices, computer networks, and basic security mechanisms
- Use professional productivity tools (Google Workspace) for academic and professional tasks
- Present and defend technical concepts related to computer fundamentals

COURSE CONTENT

- Data vs. information and basic arithmetic logic
- Repetition structures (while-wend, for-next) and logical error tracing
- Conditional statements and dataset-based problem solving
- Introduction to computer systems, number systems, and coding schemes
- Logic gates, numeric data representation, ASCII, Unicode
- Microcomputer fundamentals, processors, memory, and bus organization
- Flowchart design techniques and logical flow correction
- Input and output devices and peripheral systems
- Memory organization: primary, secondary, and solid-state storage
- Programming language basics and classification
- Computer networks (LAN, WAN, MAN) and networking devices
- Computer security concepts including malware and antivirus systems
- Practical use of Google Workspace tools (Forms, Sites, Docs, Drive)

TEACHING & LEARNING METHODS

- Lecture presentations and video content
- Interactive discussions and storytelling sessions
- Flowcharting and logical tracing exercises
- Hands-on lab sessions using G-Suite tools
- Team-based activities and presentations

READING MATERIALS

Recommended Readings

- Computer Fundamentals and ICT (2nd Edition), DIU Press, 2017 by M. Lutfar Rahman, M. Shamim Kaiser, M. Arifur Rahman, M. Alamgir Hossain

Supplementary Readings

- Computer Fundamentals (1st Edition) by Pradeep K. Sinha & Priti Sinha

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and class participation

- Class tests and quizzes
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written examination assessing conceptual understanding and problem-solving skills

AOL201: ART OF LIVING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

The Art of Living course is designed to support students in developing a balanced, ethical, and meaningful approach to personal, academic, and professional life. The course emphasizes that success is not determined by knowledge and technical skills alone, but also by behavior, values, emotional intelligence, and self-awareness. Through guided reflection, discussion, and experiential learning, students learn to positively shape their thinking patterns, attitudes, body language, and interpersonal behavior. The course encourages self-confidence, critical thinking, empathy, questioning, and lifelong learning while nurturing universal human values. Ultimately, the course prepares students to lead peaceful, responsible, and purpose-driven lives as individuals, family members, professionals, and global citizens.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply questioning techniques, goal-setting strategies, and core values to personal and professional decision-making
- Demonstrate empathy, moral responsibility, and ethical awareness in social and professional contexts
- Integrate humanitarian qualities into daily life to promote cooperation and social harmony
- Adapt effectively to a rapidly changing world through self-awareness, resilience, and continuous learning
- Analyze situations critically and respond with emotional intelligence, equity, and responsibility

COURSE CONTENT

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

- Introduction to Art of Living and growth mindset
- Thinking out of the box and understanding subconscious patterns
- Learning to learn and unlearning limiting beliefs
- Parenting awareness, self-guardianship, and relationship building
- Etiquette, personal behavior, professionalism, and workplace culture
- Self-esteem, identity awareness, and personal development
- Essential 21st-century skills and smart work practices
- Impression management, ethical living, and healthy lifestyle choices
- Mind mapping techniques for planning and problem-solving
- Transforming failure into success through reflection and resilience
- Psychosocial support, empathy, and community engagement
- Time management, financial literacy, and overcoming procrastination
- Career planning, presentation skills, and self-expression
- Universal human values and ethical decision-making

TEACHING & LEARNING METHODS

- Interactive lectures and multimedia presentations
- Group discussions, presentations, and role-play activities
- Mind mapping and reflective exercises
- Case studies, storytelling, and experiential learning
- Peer feedback and guided self-assessment

READING MATERIALS

Recommended Readings

- Art of Living by Dr. Md. Sabur Khan and Dr. Syed Mizanur Rahman 2nd Edition (2019), Daffodil International University.

Supplementary Readings

- Art of Living Workbook By Dr. Md. Sabur Khan and Dr. Syed Mizanur Rahman.
- Nash, Robert J., Answering the Virtue crats: A Moral Conversation on Character Education, New York, Teachers College Press.
- Lickona, Thomas, Education for Character: How our Schools can Teach Respect and Responsibilities, New York, Bantam Books,1991.
- Fred Newman and Phyllis Goldberg, Performance of a Lifetime: A Practical Philosophical guide to the Joyous Life. Castillo International Inc, 1996. John S. Mackenzie, A Manual of Ethics, Surjeet Publication June 2004.
- Dr. Ranjit Singh Malhi. Make Yourself Employable.
- Islam, Yousuf M, Mind Mapping and Student Organizer (Teaching Materials), HRDI-DIU 2012.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation

- Quizzes and class tests
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating understanding, application, and reflection

MCT137: PERSPECTIVE STUDY

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT113

TEACHING OBJECTIVES

The Perspective Studies course is designed to build a strong foundational understanding of perspective drawing as a core visual communication skill for disciplines such as animation, graphic design, architecture, interior design, and fine arts by integrating theoretical principles with freehand practices of one-point, two-point, and three-point perspective systems. Through structured studio exercises, observational drawing, and outdoor sketching, students learn to accurately represent three-dimensional space on a two-dimensional surface while developing an understanding of visual perception, eye level, spatial relationships, and geometric construction. By engaging with geometric forms, natural elements, architectural structures, landscapes, and real-world environments, the course encourages observational accuracy, creative interpretation, and critical analysis of form, depth, scale, and proportion, ultimately preparing students to confidently apply perspective drawing skills in academic, professional, and creative contexts.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply theoretical and practical principles of one-point, two-point, and three-point perspective in freehand drawing
- Demonstrate an understanding of eye level, vanishing points, horizon lines, and spatial depth
- Construct accurate three-dimensional geometric and organic forms on a two-dimensional plane
- Analyze spatial relationships between objects from different viewpoints and distances
- Produce perspective drawings of natural elements, architectural forms, interiors, exteriors, and landscapes

- Integrate observation, creativity, and technical accuracy in final perspective-based artworks

COURSE CONTENT

- Introduction to perspective drawing and visual perception
- Principles of eye level, horizon line, and vanishing points
- One-point, two-point, and three-point perspective systems
- Drawing three-dimensional geometric forms (cube, cylinder, cone)
- Freehand perspective techniques using pencil, pen, ink, and mixed media
- Understanding scale, proportion, and viewer position
- Perspective drawing of natural elements and organic forms
- Outdoor observational drawing and environmental perspective studies
- Architectural perspective including buildings, interiors, exteriors, staircases, and bridges
- Perspective representation of heritage sites and urban structures
- Landscape and riverscape perspective composition
- Atmospheric perspective and spatial depth in landscapes
- Final project planning, development, and refinement
- Presentation of perspective-based final artwork

TEACHING & LEARNING METHODS

- Interactive studio demonstrations and guided drawing sessions
- Lecture slide presentations and instructional videos
- Brainstorming sessions and open discussions
- Outdoor drawing and observational sketching
- Use of drawing toolkits and traditional media
- Critique sessions, peer feedback, and instructor evaluation
- Gamified learning activities and interactive platforms

READING MATERIALS

Recommended Readings

- D'Amelio, Joseph. Perspective Drawing Handbook.

Supplementary Readings

- Powell, William F. The Art of Basic Drawing, Walter Foster Creative Team
- Kessler, Peter & Uderzo, Albert. The Complete Guide to Asterix (The Adventures of Asterix and Obelix)

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance and participation
- Lab reports and sketch submissions
- Lab performance and ongoing progress

- Final practical examination / project
-

ENG113: BASIC FUNCTIONAL ENGLISH AND ENGLISH SPOKEN

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

This course is designed for first-year undergraduate students studying English in an EFL context with intermediate to low-advanced proficiency, aiming to strengthen their functional, academic, and communicative competence in English. The course focuses on developing listening and speaking skills alongside essential grammatical accuracy, enabling students to communicate effectively in both formal and informal academic and real-life situations. Through interactive classroom activities, technology-enhanced learning, and structured practice aligned with IELTS listening and speaking formats, students gain confidence in oral communication, improve comprehension skills, and develop foundational writing abilities necessary for university-level study and global communication contexts.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Communicate effectively in English in academic and everyday contexts
- Demonstrate improved listening and speaking skills using appropriate strategies
- Apply grammatical structures accurately in spoken and written communication
- Use technology efficiently for learning and practicing the English language
- Perform confidently in IELTS-style listening and speaking tasks

COURSE CONTENT

- Introduction to functional English and spoken communication
- Listening for gist and listening for specific information
- Listening for detailed understanding, tone, attitude, and main ideas
- Note-taking strategies from spoken texts
- Self-introduction techniques and video résumé preparation
- Role-play, group discussions, and interactive speaking activities
- Introduction to IELTS listening and speaking formats (Parts I & II)

- Practice sessions using IELTS-style mock tests (listening, speaking, reading, writing)
- Basic grammatical categories: nouns, articles, prepositions, adverbs, adjectives
- Sentence patterns and sentence structure (simple, compound, complex)
- Tenses, subject–verb agreement, conditionals, and transformations
- Error correction and common mistakes in English usage
- Use of technology for English language learning and communication
- Student presentations, feedback sessions, and performance reviews
- Review classes and final examination preparation

TEACHING & LEARNING METHODS

- Interactive lectures using multimedia and digital platforms
- Listening and speaking practice through audio, video, and real-life tasks
- Role-play, group discussions, and presentations
- Grammar-focused exercises and guided practice
- Technology-supported learning and online resources
- Continuous feedback, peer evaluation, and instructor-led review sessions

READING MATERIALS

Recommended Readings

- Eastwood, John. A Basic English Grammar: Exercises
- Jones, Leo. Grammar Practice Activities for Intermediate Students of English
- Jupp, T. C. & Milne, John. Guided Paragraph Writing
- Cambridge IELTS series
- Selected online learning resources

Supplementary Readings

- Murphy, Raymond. Essential English Grammar
- Fitikides, T. F. Common Mistakes in English

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating listening comprehension, grammar, application, analysis, and communicative competence

ENG123: WRITING AND COMPREHENSION

Course Type	: Theory Lab
Credits	: 1 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: ENG113

TEACHING OBJECTIVES

Writing and Comprehension is designed to develop first-year undergraduate students' academic reading and writing proficiency in an EFL context, with particular emphasis on skills required for university study and professional communication. The course focuses on strengthening reading comprehension strategies, grammatical accuracy, and structured writing through guided practice in letters, emails, essays, and academic texts. By integrating IELTS-oriented reading and writing tasks, technology-enhanced learning, and discipline-relevant materials tailored to students of Science, Information, and Technology, the course aims to enhance students' ability to process written information critically, express ideas coherently, and communicate effectively in academic, professional, and real-life contexts.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Identify key information from texts using skimming and scanning techniques
- Apply reading strategies for academic and general-purpose texts
- Use grammatical structures accurately in written English
- Communicate ideas effectively through structured written texts
- Prepare and present a professional video résumé
- Organize curriculum vitae and cover letters for employment purposes

COURSE CONTENT

- Overview of basic English communication skills
- Reading techniques: skimming, scanning, and reading for detail
- Reading comprehension tasks: True/False/Not Given, flow charts, MCQs, matching headings, matching information, short-answer questions
- Guessing word meanings from context
- Review and mock tests for reading subskills
- Developing writing syntax: sentence fragments, run-on sentences
- Grammar practice: modifiers, sentence clarity, and accuracy
- Structure and format of informal and formal letters
- Email writing for academic and professional contexts
- Academic essay writing: structure, types, and development (IELTS Writing Task 2)
- Drafting, revising, and finalizing academic essays

- Integrated reading and writing practice sessions
- IELTS-oriented mock tests for reading and writing
- CV, cover letter, and video résumé preparation
- Peer review, feedback sessions, and revision workshops
- Final review sessions and examination preparation

TEACHING & LEARNING METHODS

- Interactive lectures using multimedia and digital platforms
- Guided reading and writing practice in class and language labs
- Individual and group writing tasks and peer review activities
- Mock tests and exam-oriented practice sessions
- Technology-assisted learning using online platforms and applications
- Continuous feedback, presentations, and instructor-led discussions

READING MATERIALS

Recommended Readings

- English II Course Reader for Information and Technology

Supplementary Readings

- English for Academic Purposes 1 & 2 – British Council and UGC
- Langan, John. English Skills
- IELTS Online Practice Resources
- Grammarly, Duolingo, and similar language-learning applications

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and writing tasks
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Written assessment measuring reading comprehension, writing accuracy, organization, analysis, and application of language skills

MCT136: COMPUTER NETWORKS AND INTERNET

Course Type	: Theory Lab
Credits	: 13
Duration	: One semester

Total Contact Hours : 40

Prerequisite(s) : None

TEACHING OBJECTIVES

The Computer Networks and Internet course is designed to introduce first-year students to the fundamental concepts, principles, and technologies that enable data communication and internet-based services. The course aims to develop a clear understanding of how computers communicate over local and global networks, how the Internet functions, and how networking supports modern digital applications and services. Through conceptual explanations and practical examples, students learn about network models, protocols, transmission media, addressing schemes, and basic network security. Emphasis is placed on building a strong conceptual foundation, technical vocabulary, and analytical thinking skills necessary for advanced studies in computing, multimedia, and information technology.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Explain the basic concepts and components of computer networks
- Describe the architecture and working principles of the Internet
- Identify different network types, topologies, and transmission media
- Understand common networking protocols and communication models
- Recognize basic network security issues and safe Internet practices

COURSE CONTENT

- Introduction to computer networks and data communication
- Evolution of computer networks and the Internet
- Network components: clients, servers, switches, routers, and hubs
- Types of networks: LAN, MAN, WAN, PAN, and WLAN
- Network topologies and architectures
- Transmission media: guided and unguided media
- Data transmission concepts and signal types
- Network models: OSI and TCP/IP models
- Functions of different network layers
- Internet protocols: IP, TCP, UDP, HTTP, HTTPS, FTP, SMTP, and DNS
- IP addressing and basic subnet concepts
- Domain names and URL structure
- Internet services and applications
- Web technologies and cloud-based services
- Introduction to network security concepts
- Threats, malware, firewalls, and basic protection methods
- Ethical and responsible use of the Internet
- Review and consolidation of concepts

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations
- Conceptual discussions and real-life examples
- Demonstrations of networking concepts and tools
- Problem-solving and guided practice sessions
- Continuous feedback and question–answer sessions

READING MATERIALS

Recommended Readings

- Forouzan, Behrouz A. Data Communications and Networking. McGraw-Hill.

Supplementary Readings

- Kurose, James F., and Ross, Keith W. Computer Networking: A Top-Down Approach. Pearson.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written examination evaluating understanding of networking concepts, protocols, and Internet fundamentals

CIS112: MATHEMATICS-I FUNDAMENTALS OF MATHEMATICS

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

Mathematics I is designed to provide first-year engineering students, particularly those pursuing textile engineering, with a strong foundation in differential and integral calculus

necessary for solving real-world engineering problems. The course emphasizes the concepts of functions, limits, continuity, differentiation, and integration, focusing on both algebraic and transcendental functions of a single variable. Through systematic development of analytical techniques and problem-solving strategies, students learn to model engineering situations, analyze rates of change, and apply calculus-based methods with or without the use of technology, thereby building essential mathematical competence for advanced engineering studies.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Recognize properties, classifications, and types of functions
- Sketch graphs of functions using transformations and analytical techniques
- Evaluate limits using basic rules and L'Hospital's rule
- Apply differentiation techniques to algebraic, implicit, and logarithmic functions
- Solve applied problems involving related rates, maxima, and minima

COURSE CONTENT

- Introduction and overview of the course structure and objectives
- Concept of functions and classification of various types of functions
- Graphical representation and sketching of functions
- Limits and continuity of functions
- Differential coefficients and rate of change
- Differentiation of algebraic and transcendental functions
- Successive differentiation and Leibniz's theorem
- Applications of differentiation in optimization problems
- Maxima and minima of functions
- Introduction to partial differentiation
- Integration: definition, indefinite integrals, and definite integrals
- Applications of integration in engineering problems
- Review sessions, quizzes, presentations, and examination preparation

TEACHING & LEARNING METHODS

- Interactive lectures with problem-solving demonstrations
- Step-by-step analytical derivations and guided practice
- Classroom discussions and worked examples
- Assignments and presentations to reinforce concepts
- Continuous assessment through quizzes and tests
- Review sessions before mid-term and final examinations

READING MATERIALS

Recommended Readings

- Anton, Howard; Bivens, Irl; Davis, Stephen. Calculus

Supplementary Readings

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

- Differential Calculus by B.C. Das and B.N. Mukherjee.
- Integral Calculus by B.C. Das and B.N. Mukherjee.
- Advanced Calculus-I by Dr. F. Rahman, Elius, Hafiz, and Modul.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating conceptual understanding, application, analysis, and problem-solving skills based on Bloom's taxonomy
-

CSE122: STRUCTURED PROGRAMMING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CSE112

TEACHING OBJECTIVES

Structured Programming is designed to introduce students to the fundamental principles of programming using a structured and systematic approach, forming a strong foundation for understanding modern programming languages and software development practices. The course emphasizes problem formulation, logical solution design, implementation, debugging, testing, and documentation using a high-level structured programming language (C). Through progressive learning of programming constructs, data organization techniques, and modular program design, students develop analytical thinking, problem-solving skills, and the ability to write syntactically and semantically correct programs applicable to real-world computing problems.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Store, manipulate, and process data using variables, operators, and library functions
- Design logical solutions using conditional and looping control structures
- Develop modular programs using functions, parameters, and return values

- Manage and organize large volumes of data using arrays, strings, structures, and pointers

COURSE CONTENT

- Introduction to programming concepts and structured programming principles
- Overview of C programming language and program structure
- Variables, data types, declarations, and input/output operations
- Operators and expressions: arithmetic, relational, logical, assignment, increment, and decrement
- Mathematical functions and expression evaluation
- Decision-making constructs: if, if-else, nested if-else, else-if ladder, and switch statement
- Looping constructs: for, while, and do-while loops
- Use of break and continue statements
- Arrays: declaration, initialization, and usage
- User-defined functions and modular programming
- Strings and string handling functions
- Structures and structured data organization
- Pointers and basic memory management concepts
- Program testing, debugging, and optimization
- Review sessions, quizzes, mid-term, and final examinations

TEACHING & LEARNING METHODS

- Interactive lectures using slide presentations and instructional videos
- Open discussions and guided problem-solving sessions
- Hands-on programming practice using development tools
- Gamified learning activities and interactive platforms
- Continuous feedback, Q&A sessions, and peer interaction
- Assignments and programming exercises to reinforce concepts

READING MATERIALS

Recommended Readings

- Balagurusamy, E. Programming in ANSI C (5th Edition or later)
- Schildt, Herbert. Teach Yourself C (3rd Edition or later)

Supplementary Readings

- Kanetkar, Yashavant. Let Us C
- Ritchie, Dennis. Programming in C

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes

- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating understanding, application, analysis, and program design skills based on Bloom's taxonomy

CSE123: STRUCTURED PROGRAMMING LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CSE112

TEACHING OBJECTIVES

The Structured Programming Lab is designed to provide students with extensive hands-on experience in implementing structured programming concepts through practical problem-solving using the C programming language. The course focuses on translating logical solutions into working programs, reinforcing programming discipline, debugging skills, and correct program execution. Through guided laboratory exercises and supervised practice, students develop confidence in writing, testing, and refining programs while gaining practical exposure to programming tools, compilation processes, runtime error handling, and program documentation. The lab emphasizes accuracy, efficiency, and clarity in program development, preparing students for advanced programming and real-world computing tasks.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Write, compile, execute, and debug structured programs using C
- Apply control structures to solve computational problems effectively
- Develop modular programs using user-defined functions
- Implement data storage and manipulation using arrays, strings, and structures
- Use pointers for basic memory access and data handling
- Analyze program output and correct logical and runtime errors

COURSE CONTENT

- Familiarization with programming environment, compiler, and IDE
- Writing and executing basic C programs
- Input/output handling through practical coding tasks
- Program development using arithmetic and logical expressions

- Implementation of decision-making logic through coding exercises
- Loop-based problem solving using iterative constructs
- Debugging techniques and error identification
- Practical use of arrays for data storage and processing
- Function implementation and modular program design
- String manipulation through built-in string handling functions
- Practical use of structures for organized data representation
- Introduction to pointer-based programming through simple applications
- Code optimization, formatting, and documentation practices
- Lab-based problem-solving assignments
- Review sessions and practical examination preparation

TEACHING & LEARNING METHODS

- Instructor-led demonstrations followed by supervised lab practice
- Individual and pair programming exercises
- Problem-based learning through lab assignments
- Continuous feedback during coding sessions
- Debugging workshops and error-analysis activities
- Performance-based assessment through hands-on tasks

READING MATERIALS

Recommended Readings

- Balagurusamy, E. Programming in ANSI C (Lab-oriented sections)
- Schildt, Herbert. Teach Yourself C (Practice examples)

Supplementary Readings

- Kanetkar, Yashavant. Let Us C (Programming exercises)
- Online C programming documentation and compiler manuals

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Lab attendance and participation
- Lab performance and coding efficiency
- Practical assignments and problem-solving tasks
- Mid-term practical test
- Final practical examination assessing program design, correctness, and execution

MCT222: 3D DESIGN AND MODELING LAB

Course Type : Lab

Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

The 3D Design and Modeling Lab is designed to provide intensive hands-on training in three-dimensional modeling, scene development, texturing, lighting, and rendering using Autodesk Maya. The course aims to reinforce theoretical knowledge through practical application, enabling students to gain confidence in navigating the software interface, executing professional modeling workflows, and solving real-world design problems. Emphasis is placed on asset creation accuracy, clean topology, workflow efficiency, and visual quality. Through structured lab exercises and project-based practice, students develop the ability to create complete 3D assets and scenes suitable for animation, visualization, and interactive media production.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Create 3D models using primitive, polygon, and NURBS modeling techniques
- Apply appropriate topology principles for clean and efficient models
- Develop and apply materials and textures to 3D assets
- Set up basic lighting and camera systems for scene presentation
- Render high-quality still images and simple animated outputs

COURSE CONTENT

- Introduction to Autodesk Maya lab environment and workflow
- Viewport navigation and scene organization
- Primitive-based modeling exercises
- Polygon modeling using vertex, edge, and face operations
- Extrude, bevel, bridge, and edge loop techniques
- Boolean operations and mesh cleanup
- Topology fundamentals and edge flow management
- NURBS and curve-based modeling exercises
- Lathe, loft, and revolve modeling techniques
- Modeling real-world objects and props
- Introduction to sculpting and mesh refinement
- UV mapping and UV layout fundamentals
- Applying materials and textures using Arnold
- Photoshop-based texture application
- Basic lighting setup for product and environment scenes
- Camera setup and framing for visual presentation

- Rendering still images using Arnold renderer
- Introduction to simple animation and playblast workflow
- Scene optimization and file management
- Project-based asset and scene development
- Final project submission and practical examination

TEACHING & LEARNING METHODS

- Demonstration-based laboratory sessions
- Hands-on practice using Autodesk Maya
- Guided step-by-step modeling exercises
- Project-based learning and asset development
- Individual and group critique sessions
- Continuous instructor feedback and troubleshooting support

READING MATERIALS

Recommended Readings

- Murdock, Kelly L. Autodesk 3ds Max 2019 Complete Reference Guide.

Supplementary Readings

- Ahmed, Arif. Physics for Animation.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance
- Lab reports and documentation
- Lab performance and practical execution
- Final project submission and practical examination

MCT223: MATHEMATICS-II COORDINATE GEOMETRY

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CIS112

TEACHING OBJECTIVES

Mathematics-II is designed to extend students' mathematical foundation by introducing linear algebra, complex variables, and advanced topics in differential and integral calculus that are

essential for science, engineering, and technology disciplines. The course aims to develop analytical and problem-solving skills through systematic study of matrices, vector spaces, linear transformations, and eigenvalues, alongside applications of calculus to technical and real-world problems. By integrating theoretical understanding with computational techniques, the course equips students with mathematical tools necessary for advanced coursework and interdisciplinary applications in engineering, physics, economics, and the natural sciences.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Perform integration techniques to evaluate definite and multiple integrals
- Explain fundamental concepts of matrix theory and vector algebra
- Apply matrix operations to solve systems of linear equations
- Perform basic operations and solve equations involving complex numbers
- Solve integration problems using substitution, integration by parts, and partial fractions

COURSE CONTENT

- Introduction and overview of Mathematics-II
- Partial differentiation: concepts, successive differentiation, and applications
- Definite integrals and multiple integration
- Gamma and Beta functions
- Complex numbers and complex-valued functions
- Matrices: definition, types, and basic operations
- Elementary transformations of matrices
- Solution of systems of linear equations
- Vector spaces and their properties
- Linear transformations and their applications
- Eigenvalues and eigenvectors
- Review sessions for mid-term and final examinations
- Quizzes, mid-term examination, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures supported by worked examples
- Step-by-step problem-solving demonstrations
- Classroom discussions and guided practice sessions
- Assignments and presentations to reinforce conceptual understanding
- Continuous assessment through quizzes and tests
- Review and revision sessions before examinations

READING MATERIALS

Recommended Readings

- Spiegel, Murray. Complex Variables
- Anton, Howard. Calculus (9th Edition), Wiley
- Lipschutz, Seymour. Linear Algebra

Supplementary Readings

- College Mathematics, Schaum's Outline Series (2nd Edition), McGraw-Hill

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating conceptual understanding, application, analysis, and problem-solving ability in calculus, linear algebra, and complex variables
-

GED201: BANGLADESH STUDIES

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

Bangladesh Studies is a multidisciplinary course designed to provide students with a comprehensive understanding of the historical, political, economic, social, cultural, and environmental dimensions of Bangladesh. The course aims to develop students' factual knowledge and analytical abilities by examining the inevitability of the emergence of Bangladesh as a nation-state, its journey through governance and democratization, and its contemporary development challenges. Through critical inquiry and evidence-based analysis, students are encouraged to appreciate the country's past and present realities, cultivate civic awareness, and develop problem-solving perspectives necessary to face the challenges of the twenty-first century both as informed citizens and responsible professionals.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Conduct SWOT analysis of the geography, demography, society, culture, governance, and politics of Bangladesh
- Critically analyze national and international factors influencing Bangladesh's socio-economic development

- Debate historical, legal, and ethical issues affecting contemporary Bangladesh using empirical evidence
- Explain major environmental challenges and climate change impacts from a Bangladesh perspective
- Propose rational, innovative, and sustainable solutions to key national challenges through effective communication

COURSE CONTENT

- Introduction to Bangladesh Studies: course scope, methodology, and relevance
- Bangladesh at a glance: geography, natural resources, climate, population, and demographic trends
- Social structure, culture, traditions, festivals, and national identity
- Origin of the name Bangladesh and identity of the Bengali people
- Origin and development of the Bangla language
- Bangladesh in international affairs and foreign policy principles
- Constitution of Bangladesh: formation, features, principles, rights, and amendments
- Organs of government: structure, functions, governance challenges, and e-governance
- Historical background: Language Movement and political discrimination (1947–1969)
- Six-Point Movement, 1970 election, and events leading to the Liberation War
- Liberation War of 1971: key events, Mujibnagar Government, MuktiBahini, and international role
- Economic profile of Bangladesh: development strategies, national budget, and emerging economy
- Urbanization and rural development: trends, challenges, and development models
- Environmental degradation and climate change in Bangladesh
- Industrial development with emphasis on the ICT sector
- Review sessions, presentations, quizzes, mid-term, and final examinations

TEACHING & LEARNING METHODS

- Interactive lectures supported by multimedia presentations
- Brainstorming sessions and guided discussions
- Case studies and content analysis
- Peer learning and group-based activities
- Student presentations and review classes
- Continuous feedback and reflective discussions

READING MATERIALS

Recommended Readings

- Rahman, Sheikh Mujibur. The Unfinished Memoirs
- The Constitution of the People's Republic of Bangladesh, Ministry of Law, Justice and Parliamentary Affairs

- Ahmed, Emajuddin (ed.). Foreign Policy of Bangladesh: A Small State's Imperative
- Sobhan, Rehman. Bangladesh: Problems of Governance
- Van Schendel, Willem. A History of Bangladesh
- Bangladesh Economic Review, Ministry of Finance, Government of Bangladesh

Supplementary Readings

- Mascarenhas, Anthony. Bangladesh: A Legacy of Blood
- Guhathakurta, Meghna & van Schendel, Willem (eds.). The Bangladesh Reader: History, Culture, Politics
- Selected Bangla-language historical and political texts

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating conceptual understanding, critical analysis, application, and synthesis of knowledge related to Bangladesh

MCT231: OBJECT ORIENTED PROGRAMMING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CSE122

TEACHING OBJECTIVES

Object Oriented Programming is designed to introduce students to advanced programming concepts and software design principles using a high-level object-oriented programming language such as Python or Java. The course focuses on understanding how complex software systems are modeled, designed, and implemented through object-oriented paradigms, emphasizing abstraction, modularity, reusability, and maintainability. By analyzing structured and semi-structured problem statements, students learn to identify objects, design class hierarchies, and organize code into scalable architectures. The course aims to bridge foundational programming skills with real-world software development practices, enabling students to design, implement, and evaluate reusable and extensible software solutions.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Develop object-oriented programs that integrate data and behavior effectively
- Analyze problem statements to model objects and define software architecture
- Apply object-oriented programming techniques using multiple interacting classes
- Translate abstract concepts into concrete class designs and implementations
- Present and defend technical decisions related to object-oriented software development

COURSE CONTENT

- Introduction to object-oriented programming and software engineering principles
- Code reuse, extensibility, and maintainable software design
- Review of procedural programming fundamentals
- Objects and classes: state, behavior, object identification, and references
- Class design from real-world scenarios
- Constructors, initialization techniques, and constructor chaining
- Inheritance and hierarchical class relationships
- Association, aggregation, and composition
- Abstraction, information hiding, and encapsulation
- Access modifiers and scope control
- Polymorphism: static and dynamic binding
- Abstract classes and abstract methods
- Interfaces and multiple inheritance concepts
- Exception handling and user-defined exceptions
- Object-oriented analysis of real-life scenarios and class design
- Review sessions, presentations, quizzes, mid-term, and final examinations

TEACHING & LEARNING METHODS

- Interactive lectures supported by slide presentations and instructional videos
- Brainstorming sessions and open discussions
- Case studies and real-life problem analysis
- Practical programming demonstrations and exercises
- Mind mapping for object modeling and design
- Continuous feedback through Q&A sessions and interactive platforms

READING MATERIALS

Recommended Readings

- Schildt, Herbert. The Complete Reference C#

Supplementary Readings

- Harwani, B. M. Learning Object-Oriented Programming in C#
- Oberg, Robert J. & Harkness, Howard Lee. Object-Oriented Programming in C#

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating conceptual understanding, application, analysis, and design of object-oriented software based on Bloom's taxonomy

MCT232: OBJECT ORIENTED PROGRAMMING LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CSE123

TEACHING OBJECTIVES

The Object-Oriented Programming Lab is designed to provide students with practical experience in object-oriented analysis, design, and modeling through hands-on activities that translate real-world problems into well-structured software designs. The lab emphasizes applying modern object-oriented principles such as abstraction, encapsulation, inheritance, and modularity using industry-accepted modeling techniques. Students learn to analyze problem scenarios, identify responsibilities and collaborations among objects, construct UML models, and progressively refine designs toward implementation readiness. The course strengthens design thinking, documentation skills, and best practices essential for building flexible, extensible, and maintainable object-oriented software systems.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply object-oriented analysis techniques to model real-world problem domains
- Use CRC (Class–Responsibility–Collaborator) cards to analyze and design object responsibilities
- Differentiate and apply object relationships including association, aggregation, and composition
- Create UML class, use-case, component, and deployment diagrams
- Design object-oriented software following best practices and quality principles

COURSE CONTENT

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

- Introduction to software quality, software engineering, and development life cycles
- Comparative analysis of structured vs. object-oriented approaches
- Problem and solution modeling perspectives
- Object-oriented paradigm: objects, classes, responsibilities, and collaborations
- Object-oriented analysis and design processes
- Overview of development processes (Unified Process, Agile approaches)
- Fundamentals of Unified Modeling Language (UML)
- UML notation for classes, objects, relationships, and methods
- Domain modeling and information modeling
- Structural modeling using objects, attributes, associations, links, and messages
- Use-case modeling and behavioral analysis
- Writing use-case descriptions with scenarios, extensions, and constraints
- Transition from analysis models to design models
- Design class diagrams with operations and method specifications
- Design for reuse, refactoring, and iterative improvement
- Quality-oriented design principles: cohesion, coupling, complexity, and congruence
- Physical design using UML component and deployment diagrams
- Lab-based case studies and design documentation
- Final project development, presentation, and practical examination

TEACHING & LEARNING METHODS

- Instructor-guided lab demonstrations
- Hands-on modeling and design exercises
- Case-study-based problem analysis
- Group discussions and design critiques
- UML tool usage and diagramming practice
- Continuous feedback, peer review, and design refinement sessions

READING MATERIALS

Recommended Readings

- Schildt, Herbert. The Complete Reference C#
- Gamma, Erich et al. Design Patterns: Elements of Reusable Object-Oriented Software

Supplementary Readings

- Harwani, B. M. Learning Object-Oriented Programming in C#
- Fowler, Martin et al. Refactoring: Improving the Design of Existing Code
- Kerievsky, Joshua. Refactoring to Patterns

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Lab attendance and participation
- Lab reports and design documentation
- Lab performance and practical exercises

- Final practical examination / project
-

CSE417: WEB ENGINEERING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CSE231

TEACHING OBJECTIVES

Web Engineering is designed to introduce students to the systematic development of modern web-based systems by applying software engineering principles to the design, implementation, deployment, and maintenance of web applications. The course emphasizes the analysis of requirements, architectural and interface design, usability, accessibility, and security of web systems while integrating client-side and server-side technologies. Through a combination of conceptual understanding and hands-on exposure to contemporary web technologies, students learn to design scalable, secure, and maintainable web applications capable of solving real-world problems in diverse domains.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Understand core concepts, principles, and methodologies of web engineering
- Design modern, responsive websites using front-end technologies
- Develop dynamic web applications using client-side and server-side programming languages
- Apply web engineering methodologies throughout the web application development lifecycle
- Present, analyze, and defend technical aspects of web-based systems

COURSE CONTENT

- Introduction to web technology and web engineering concepts
- Software engineering principles for web applications
- Internet and web fundamentals: protocols, domains, URLs, and web architecture
- Attributes and categories of web applications
- HTML fundamentals: document structure, tags, attributes, tables, and layouts
- CSS fundamentals: selectors, box model, positioning, typography, and styling
- Responsive web design principles
- CSS Flexbox, Grid, and media queries

- CSS frameworks with emphasis on Bootstrap
- JavaScript fundamentals: syntax, variables, control flow, functions, arrays, and DOM manipulation
- jQuery basics and asynchronous programming with AJAX
- PHP fundamentals and server-side scripting
- Object-oriented programming concepts in PHP
- Relational database interaction using PHP with MySQL (MySQLi and PDO)
- CRUD operations and database-driven web applications
- MVC architecture and introduction to Laravel framework
- NoSQL database concepts and MongoDB operations
- Web application usability, accessibility, and security considerations
- Review sessions, quizzes, mid-term, and final examinations

TEACHING & LEARNING METHODS

- Interactive lectures supported by slide presentations and instructional videos
- Brainstorming sessions and open discussions
- Case studies and real-world application analysis
- Practical demonstrations and guided exercises
- Mind mapping and interactive learning tools
- Continuous feedback through quizzes, assignments, and presentations

READING MATERIALS

Recommended Readings

- Kappel, Gerti et al. Web Engineering: The Discipline of Systematic Development of Web Applications

Supplementary Readings

- Nixon, Robin. Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and project-based tasks
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Written assessment evaluating understanding, application, analysis, and design of web-based systems based on Bloom's taxonomy

CSE418: WEB ENGINEERING LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CSE232

TEACHING OBJECTIVES

Web Engineering Lab is designed to provide students with practical, hands-on experience in developing modern, responsive, and dynamic web applications using contemporary front-end and back-end technologies. The course focuses on implementing client–server communication, designing responsive user interfaces, validating data on both client and server sides, and integrating databases to build complete web solutions. Through guided laboratory exercises and project-based learning, students gain experience in applying web technologies to solve real-world problems, developing scalable applications, and understanding the workflow of modern web systems from request handling to deployment-ready functionality.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Explain and demonstrate client–server communication mechanisms in web applications
- Design and implement responsive and user-friendly web interfaces
- Develop full-stack web applications using front-end and back-end technologies
- Implement real-time data validation, asynchronous operations, and database interactions
- Present and defend technical decisions related to modern web application development

COURSE CONTENT

- Understanding HTTP protocols and client–server interaction
- Anatomy of web requests, URLs, and HTTP methods
- Working with HTML multimedia elements and advanced form controls
- Designing interactive and styled user interfaces using CSS
- CSS transitions, transforms, and layout techniques
- Designing responsive navigation bars using pure CSS and Flexbox
- Creating responsive page layouts for mobile and tablet devices
- Building complete website layouts using HTML and CSS
- Client-side form validation using JavaScript and DOM manipulation
- Real-time error handling and asynchronous form submission using AJAX
- jQuery fundamentals and dynamic UI manipulation
- Asynchronous operations using jQuery AJAX
- PHP fundamentals for dynamic web processing

- Session management, cookies, and server-side data handling in PHP
- Server-side form validation and file upload mechanisms
- Database interaction using PHP for CRUD operations
- Implementing MVC architecture using Laravel
- Authentication, authorization, and ORM-based database operations
- Integrating MongoDB with PHP for NoSQL data management
- Mini-project development, review sessions, and final practical examination

TEACHING & LEARNING METHODS

- Instructor-led demonstrations followed by supervised lab practice
- Hands-on coding exercises and guided experimentation
- Problem-based learning through real-world lab tasks
- Case-study-driven mini projects
- Continuous feedback during development sessions
- Peer review, debugging workshops, and performance evaluation

READING MATERIALS

Recommended Readings

- Kappel, Gerti et al. Web Engineering: The Discipline of Systematic Development of Web Applications

Supplementary Readings

- Nixon, Robin. Learning PHP, MySQL & JavaScript with jQuery, CSS & HTML5

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Lab attendance and participation
- Lab reports and documentation
- Lab performance and hands-on execution
- Final practical examination / project

MCT313: 3D ANIMATION I

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT321

TEACHING OBJECTIVES

This course is designed to introduce students to the fundamental principles and practical techniques of three-dimensional animation using professional industry software, specifically Autodesk 3ds Max. The course aims to develop a clear understanding of animation workflows, keyframe animation, controller-based animation, motion dynamics, and rendering techniques required for contemporary animation and visual effects production. Through hands-on practice and guided projects, students learn how to animate objects, control motion paths, manage animation curves, and generate high-quality rendered outputs suitable for multimedia, broadcast, and cinematic applications. The course prepares students with essential animation skills aligned with industry standards and production pipelines.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Perform keyframe animation using professional 3D animation tools
- Animate objects using controllers and motion constraints
- Create looping and continuous animations using curve editors
- Apply animation controllers for position, rotation, and scale
- Implement special animation effects and motion dynamics
- Use path, look-at, and link constraints effectively
- Render still images and animation sequences for final output

COURSE CONTENT

- Introduction to animation fundamentals and 3D animation workflow
- Keyframe animation techniques using Set Key and Auto Key
- Motion control using Curve Editor for realistic movement
- Principles of acceleration, deceleration, and uniform motion
- Bouncing ball animation and basic motion studies
- Animation using controllers: path, look-at, and force-based motion
- Rendering techniques for still images and animation sequences
- Render Scene settings, output formats, and RAM Player usage
- Video Post system and post-production rendering effects
- Lens flare, glow, and visual effects rendering
- Circular path animation and banking parameters
- Copying and managing animation controllers
- Creating continuous and loop animations using curve editor settings
- Working with default animation controllers (position, rotation, scale)
- Advanced controller types: list controllers, PRS, TCB, noise, and wave controllers
- Animation constraints for camera, light, and object control
- Walkthrough assistant and scene navigation
- Sound integration and project management
- Batch rendering and scene state management
- Project presentation, review sessions, mid-term, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures with visual demonstrations
- Hands-on practical sessions using 3D animation tools
- Guided animation exercises and studio practice
- Project-based learning and industry-oriented assignments
- Critique sessions, feedback, and peer review
- Workshops and presentations aligned with production pipelines

READING MATERIALS

Recommended Readings

- Ahmed, Arif. Physics for Animation

Supplementary Readings

- Murdock, Kelly L. Autodesk 3ds Max Complete Reference Guide

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Quizzes and class tests
- Assignments and practical exercises
- Mid-term examination

Semester End Examination (40%), including:

- Practical and written assessment evaluating animation principles, controller usage, motion quality, and rendering output

MCT314: CREATIVE WRITING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

This course is designed to unlock students' imaginative potential and develop their ability to produce original, meaningful, and purpose-driven written content for creative and professional contexts. The course focuses on concept development, mind mapping, character creation, storytelling techniques, dialogue writing, and narrative structure across fiction, non-fiction,

poetry, and multimedia-oriented writing. Through intensive practice, exercises, and critical discussion, students learn to transform ideas into compelling stories, scripts, and creative briefs suitable for animation, film, games, advertising, and digital media. The course emphasizes clarity of expression, emotional engagement, cultural awareness, and narrative coherence while preparing students to build a professional writing portfolio and explore publication opportunities.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Develop creative concepts through mind mapping, research, and character construction
- Construct stories using effective plot development, scene design, and dialogue
- Apply imagination, cultural awareness, and human psychology in creative writing
- Understand creative content production pipelines for multimedia applications
- Apply storyboarding techniques for visual and narrative-based media projects

COURSE CONTENT

- Introduction to creative writing and its applications
- Difference between general writing and creative writing
- Creative writing fields and professional opportunities
- Idea generation, mind mapping, and creative sketching
- Character creation, background development, and character arcs
- Human behavior, psychology, culture, and social influences in writing
- Brainstorming techniques and maintaining idea journals
- Writing effective dialogue and managing conflict
- Developing creative briefs for branding and media projects
- Research techniques for creative writing
- Conveying characters' inner thoughts, emotions, and body language
- Storytelling techniques and story analysis sessions
- Sensory writing: sight, sound, smell, taste, and texture
- Use of metaphors, similes, and symbolism
- Plot development, structure, and narrative flow
- Openings, climaxes, resolutions, and thematic consistency
- Writing for children and understanding child psychology
- Managing time in narratives, flashbacks, and timelines
- Structuring longer works: chapters, scenes, and subplots
- Building suspense, tension, and pacing
- Scriptwriting, storyboarding, and concept art fundamentals
- Portfolio development, presentations, quizzes, mid-term, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures and storytelling sessions
- Brainstorming, mind mapping, and creative workshops
- Writing exercises, peer review, and critique sessions

- Outdoor and experiential learning activities
- Short film screenings and narrative analysis
- Project-based learning and portfolio development
- Continuous feedback, discussions, and presentations

READING MATERIALS

Recommended Readings

- Brand, William. Creative Writing Exercises for Dummies

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and creative writing tasks
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Evaluation of creative work, narrative development, originality, structure, and application of storytelling techniques

MCT315: 3D ENVIRONMENT DESIGN

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT321

TEACHING OBJECTIVES

3D Environment Design is designed to introduce students to the principles, tools, and workflows involved in creating immersive and interactive three-dimensional virtual environments using industry-standard software such as VUE and Autodesk 3ds Max. The course focuses on both the technical and aesthetic aspects of digital environment creation, including terrain generation, vegetation systems, atmospheric effects, lighting, and photorealistic rendering. By exploring real-time and pre-rendered environment pipelines, students gain practical experience in building visually compelling and functionally coherent virtual worlds suitable for animation, games, architectural visualization, and interactive media,

while developing an understanding of the design considerations unique to large-scale virtual environments.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Design and construct digital 3D environments using professional software tools
- Create and manage environmental elements such as terrain, vegetation, water, and rocks
- Animate natural elements including trees, grass, and clouds
- Apply lighting and atmospheric effects to enhance realism
- Produce photorealistic environment renders and animated outputs

COURSE CONTENT

- Introduction to 3D environment creation tools: VUE, 3ds Max, and Maya
- Navigating the VUE interface and basic object manipulation
- Working with infinite planes and scene navigation
- VUE workflow fundamentals: primitive objects and terrain editor
- Creating plants, vegetation, and ecosystem elements
- Modeling rocks, meta clouds, planets, and metaballs
- Grouping, Boolean operations, and scene organization
- Linking and targeting objects within the environment
- Lighting setup and light editor usage
- Scene alignment, smart drop tools, and wireframe selection
- Color management, render options, and animation rendering
- Creating complete environments and exporting outputs
- Advanced effects: atmosphere editor and material editor
- Ecosystem painting and advanced vegetation control
- Linking VUE environments with 3ds Max assets
- Creating architectural structures in 3ds Max and integrating with VUE
- Photorealistic scene development and final project execution
- Review sessions, quizzes, mid-term, presentations, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures supported by visual demonstrations
- Hands-on studio sessions using 3D environment tools
- Guided practical exercises and workflow demonstrations
- Project-based learning and environment design assignments
- Critique sessions, feedback, and peer review
- Workshops aligned with industry environment pipelines

READING MATERIALS

Recommended Readings

- Chopine, Ami & Chopine, Vladimir. Vue: From the Ground Up – The Official Guide

Supplementary Readings

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

- Ahmed, Arif. Physics for Animation

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Quizzes and class tests
- Assignments and environment design exercises
- Mid-term examination

Semester End Examination (40%), including:

- Practical and written assessment evaluating environment design quality, realism, animation, and rendering output

CIS332: HUMAN COMPUTER INTERACTION

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

Human–Computer Interaction is designed to introduce students to the principles, theories, and practices involved in designing effective, usable, and meaningful interactions between humans and digital systems. The course focuses on understanding human psychology, cognition, perception, memory, culture, and social behavior, and how these factors influence user-centered interface and experience design. Through a combination of theoretical foundations, real-life case studies, design thinking methodologies, and practical UI/UX tools, students learn to analyze user needs, define usability and experience goals, design intentional interfaces, and evaluate interactive systems. The course prepares students to make informed design decisions and to develop industry-ready UI/UX project deliverables.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Analyze human psychology, demographic factors, culture, and social constraints relevant to interaction design
- Explain human memory structures, cognitive processes, and technological limitations
- Apply human-centered design concepts and design thinking methodologies
- Use UI/UX design principles and professional tools for interface design

- Collaborate on real-life projects and develop interactive prototypes

COURSE CONTENT

- Introduction to Human–Computer Interaction and digital interaction paradigms
- Evolution of digital technology and the importance of UI/UX
- Human psychology, behavior, culture, demographics, and social factors
- Seven stages of action and levels of cognitive processing
- Human-centered design philosophy and usability goals
- Storytelling as a design and problem-framing tool
- Knowledge in the head vs. knowledge in the world
- Human memory structure and mental models
- Natural mapping, cultural influence on design, and trade-offs in interaction
- Constraints, affordances, signifiers, feedback, and discoverability
- Human errors: slips, mistakes, and error prevention strategies
- Design thinking and problem-solving frameworks (Double Diamond model)
- Good design vs. bad design and ethical design considerations
- Visual and UX design principles: layout, color, typography, unity, and hierarchy
- Responsive design concepts and wireframing techniques
- UI/UX design tools and prototyping workflows
- User research, empathy mapping, and user journey analysis
- Use-case diagrams, functional workflows, and storyboarding
- Prototype testing, evaluation, and iteration
- Group-based real-life project development and presentation
- Review sessions, quizzes, mid-term, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures and multimedia presentations
- Brainstorming sessions and real-life case studies
- Group discussions and storytelling activities
- Design thinking workshops and project grooming sessions
- Practical UI/UX tool demonstrations and hands-on exercises
- Continuous feedback, critique, and peer review

READING MATERIALS

Recommended Readings

- Norman, Don. The Design of Everyday Things

Supplementary Readings

- Dix, Alan; Finlay, Janet. Human–Computer Interaction (3rd Edition)
- Cabrera, James. Modular Design Framework: A Project-Based Guide for UI/UX Designers

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and design exercises
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Written and design-oriented assessment evaluating understanding, application, analysis, and creation of human-centered interactive systems

MCT321: COMPUTER AIDED DESIGN

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	:

TEACHING OBJECTIVES

The Computer Aided Design (CAD) course is designed to develop students' technical proficiency and conceptual understanding of digital drafting using industry-standard CAD software. The course familiarizes students with fundamental and advanced CAD tools for creating precise 2D drawings used in engineering, architecture, product design, and manufacturing. Through structured instruction and hands-on practice, students learn drafting standards, geometric construction, layering, dimensioning, tolerances, and annotation techniques while developing accuracy, efficiency, and problem-solving skills. Emphasis is placed on translating design concepts into technically correct 2D digital drawings that meet professional and industrial standards.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Create accurate and detailed 2D technical drawings using CAD software
- Apply standard drafting conventions, dimensions, annotations, and symbols
- Use geometric construction techniques and precision tools effectively
- Interpret and produce working drawings suitable for manufacturing or construction
- Demonstrate efficient CAD workflows, layering systems, and file management practices

COURSE CONTENT

- Introduction to Computer Aided Design and CAD applications.
- Overview of CAD software interface and workspace customization.
- Coordinate systems, units, and drawing setup.
- Basic drawing tools including lines, circles, arcs, polygons, splines, and polylines.
- Editing tools such as trim, extend, offset, fillet, chamfer, mirror, scale, and array.
- Object snaps, grids, layers, and precision control.
- Annotation tools including text, dimensions, leaders, hatching, and symbols.
- Drafting standards and technical drawing conventions.
- Geometric constraints and accurate drawing relationships.
- Creating, editing, and managing blocks and reusable components.
- Layouts, viewports, scaling, and plotting techniques.
- Importing and exporting CAD files.
- Project-based 2D drafting exercises.
- Review sessions and drawing refinement

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations
- Demonstration-based practical sessions
- Hands-on CAD exercises and assignments
- Problem-solving and design-based learning
- Guided project development
- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Omura, George. Mastering AutoCAD. Sybex.

Supplementary Readings

- Leach, Paul. Beginning AutoCAD.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating technical accuracy, application, and design competency

MCT324: 3D ANIMATION 2

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT313

TEACHING OBJECTIVES

3D Animation–2 is an advanced, practice-oriented course designed to develop students’ skills in dynamic simulation and effect-based animation using industry-standard software such as Autodesk 3ds Max and Maya. The course focuses on object dynamics, particle systems, space warps, controllers, and dynamic simulations to create realistic motion and visual effects. Through hands-on experimentation, studio exercises, and project-based learning, students gain an understanding of physical forces, particle behavior, deformation techniques, and simulation workflows essential for professional animation, visual effects, and motion design pipelines.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Create animations using particle systems and environmental forces
- Apply space warps and deformation techniques for effect-based animation
- Use controllers to generate wave motion and natural movement
- Develop particle dynamics and simulation effects in 3ds Max and Maya
- Produce complete dynamic simulation projects using object and particle dynamics

COURSE CONTENT

- Introduction to particle systems and dynamic animation principles
- Particle systems: spray, super spray, gravity, wind, deflectors
- Environmental effects: wave, ripple, bomb, snow, blizzard
- Motion controls: path follow, push, motor
- Free Form Deformation (FFD) techniques
- Effect animations such as objects passing through constraints
- Displace space warps and modifier-based warps
- Bend, noise, skew, taper, twist, and stretch modifiers
- Wave motion and waveform controllers
- Noise controllers and environmental animation effects
- Particle dynamics fundamentals
- Fountain and fluid-style particle effects in 3ds Max
- Introduction to Maya particle dynamics
- Object dynamics and collision-based animation in Maya
- Particle goals, instances, and colliders

- Complete dynamic simulation project development
- Project presentation, review sessions, and final examination

TEACHING & LEARNING METHODS

- Hands-on studio sessions in computer labs
- Demonstrations of dynamic simulation techniques
- Guided practical exercises and effect animation tasks
- Project-based learning and iterative development
- Feedback, critique sessions, and technical discussions
- Industry-oriented workflow demonstrations

READING MATERIALS

Recommended Readings

- Ahmed, Arif. Physics for Animation

Supplementary Readings

- Murdock, Kelly L. Autodesk 3ds Max Complete Reference Guide

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance and participation
- Lab reports and practical exercises
- Lab performance and dynamic animation tasks
- Final project and examination assessing technical execution, realism, and creativity

MIS575: PROJECT MANAGEMENT

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: None

TEACHING OBJECTIVES

Project Management is designed to equip students with the knowledge, skills, and professional practices required to successfully plan, execute, manage, and document academic and industrial projects in the field of Multimedia and Creative Technologies (MCT). The course focuses on both core project management principles and essential soft skills, enabling students

to handle real-world project constraints related to scope, time, cost, risk, quality, and stakeholder expectations. Through case studies, hands-on planning tools, proposal and report writing, team collaboration, and presentation activities, students are prepared to undertake their final year project confidently and to apply structured project management approaches in professional environments.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Identify and apply contemporary project management concepts in multimedia technology projects
- Demonstrate leadership and teamwork skills required for effective project management
- Develop and evaluate project proposals for academic and industrial contexts
- Plan, execute, monitor, and control multimedia projects using standard project management tools
- Produce a complete final-year project report suitable for academic defense

COURSE CONTENT

- Understanding projects in multimedia and creative technology domains
- Overview of project management concepts and core components
- Types of projects with emphasis on multimedia and technology-based projects
- Roles, responsibilities, and competencies of project managers
- Management principles and managerial functions
- Cultural, cognitive, and organizational challenges in project management
- Project life cycle models and six-phase project management frameworks
- Project scope definition, activity sequencing, and success/failure analysis
- Project planning techniques: WBS, Gantt charts, PERT, and CPM
- Budgeting, cost estimation, bidding, and resource allocation
- Risk identification, analysis, and mitigation strategies
- Decision-making models and stakeholder management processes
- Project proposal writing: structure, content, and evaluation
- Group development, team building, and collaboration strategies
- Project reporting and documentation practices
- Final-year project report structure and chapter-wise writing guidelines
- Academic referencing and APA citation style
- Project presentations, demonstrations, quizzes, mid-term, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures with real-life multimedia project examples
- Brainstorming sessions and group discussions
- Case studies and project-based learning activities
- Proposal writing workshops and report development sessions
- Student presentations and peer feedback
- Continuous mentoring and project grooming

READING MATERIALS

Recommended Readings

- Meredith, J. R., Shafer, S. M., & Mantel Jr., S. J. Project Management: A Strategic Managerial Approach (10th Edition), John Wiley & Sons
- Robbins, S. P., & Coulter, M. Management (14th Edition), Pearson
- Project Management Institute. A Guide to the Project Management Body of Knowledge (PMBOK® Guide) (6th Edition)

Supplementary Readings

- Nicholas, J. M. Project Management for Business and Technology: Principles and Practice (2nd Edition)

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and project reports
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Written and analytical assessment evaluating project planning, implementation strategies, proposal writing, and reporting skills

MCT331: VISUAL EFFECTS PRODUCTION

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT233

TEACHING OBJECTIVES

Visual Effect & Production is designed to introduce students to the principles, tools, and workflows used in modern visual effects and video post-production. The course focuses on understanding the role of VFX in film, television, advertising, and digital media, while developing practical skills using industry-standard software such as Adobe After Effects. Students learn core compositing techniques, motion graphics, tracking, masking, lighting effects, and particle-based effects, along with an understanding of production pipelines and real-footage integration. Through structured exercises and project-based learning, the course

prepares students to design, execute, and deliver professional-quality visual effects suitable for contemporary media platforms.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Navigate and use the Adobe After Effects interface effectively
- Apply tracking, masking, and keyframe animation techniques
- Create lighting, flare, glow, and particle-based visual effects
- Use fractal noise, heat distortion, and blending techniques creatively
- Composite animated elements with real footage and render final outputs

COURSE CONTENT

- Introduction to visual effects, CGI, and SFX in media production
- History and applications of visual effects in film and digital media
- Adobe After Effects interface and VFX-oriented workspace
- 2D tracking and motion tracking techniques
- 3D camera tracking and scene integration
- Layer types: solid, adjustment, null, and shape layers
- Text animation in 2D and 3D environments
- Masking techniques and path-based animation
- Advanced keyframe animation using the graph editor
- Optical flare creation and color grading techniques
- Custom stroke and 3D path animations
- Lighting effects, glow effects, and light source simulation
- Heat distortion effects for environmental realism
- Fractal noise generation and abstract effect creation
- Particle effects using Trapcode Particular
- Energy and glow effects using Saber
- Blending modes and track matte techniques
- Real footage compositing and composition management
- Exporting and rendering using After Effects and Media Encoder
- Quizzes, mid-term examination, project presentation, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures and video demonstrations
- Hands-on studio sessions using VFX software tools
- Practical exercises in compositing and motion graphics
- Outdoor shooting sessions for chroma, camera, and lighting practice
- Project-based learning and final VFX composition
- Continuous feedback, critique, and performance evaluation

READING MATERIALS

Recommended Readings

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

- Finance, Charles & Zwerman, Susan. The Visual Effects Producer: Understanding the Art and Business of VFX

Supplementary Readings

- Clark, Barbara; Spohr, Susan; Higginbotham, Dawn; Bakhru, Kumari. The Guide to Managing Postproduction for Film, TV, and Digital Distribution

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and practical VFX tasks
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Practical and written assessment evaluating compositing quality, technical execution, creativity, and production workflow

MCT332: GRAPHIC NOVEL

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT113

TEACHING OBJECTIVES

This course is designed to develop students' critical, creative, and visual storytelling abilities through the study and production of comics and graphic novels. The course traces the evolution of visual narrative from prehistoric cave drawings to modern graphic novels, emphasizing how images and text function independently and in combination to create meaning. Students are encouraged to engage in informed, nuanced, and critically reasoned debates regarding the cultural, artistic, and educational value of graphic novels while learning the foundational principles of drawing, panel composition, storytelling, and storyboarding. Through analysis, discussion, and hands-on creation, the course prepares students to apply graphic narrative techniques in animation, illustration, visual communication, and multimedia production.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Explore and analyze diverse genres of comics and graphic novels

- Explain how text and image interact to construct narrative meaning
- Critically examine distinctions between “high” and “low” art forms
- Conduct independent research and communicate arguments clearly in written form
- Present and discuss interpretive insights and creative work effectively

COURSE CONTENT

- Introduction to visual storytelling and the evolution of graphic narratives
- Cave drawings and the origins of visual communication
- Development of cartoons from stick figures to modern forms
- Types of cartoons: gag, pocket, strip, editorial, and experimental forms
- Cartoon strips and comic strips as narrative tools
- Emergence and classification of comics: general, manga, and hyper comics
- Historical development of the graphic novel
- Anatomy of graphic novels and panel analysis
- Six-panel basic structure and foundational layout principles
- Writing stories for graphic novels: structure and narrative flow
- Story endings, cliffhangers, and reader engagement techniques
- Narrative devices such as red herrings and suspense
- Comic-making techniques and visual storytelling tools
- Breaking the fourth wall in comics and graphic novels
- Use of onomatopoeia and sound representation in visual narratives
- Panel shots and camera angles in graphic storytelling
- Storyboarding principles for comics and animation
- Conceptual perspectives for narrative development
- Relationship between panels and sequential flow
- Business and professional opportunities in graphic novels
- Presentations, reviews, quizzes, mid-term, and final examination

TEACHING & LEARNING METHODS

- Interactive lectures and guided discussions
- Critical reading and debate sessions
- Brainstorming, mind mapping, and visual analysis
- Drawing and storyboarding workshops
- Short storytelling and creative writing exercises
- Project-based learning and peer critique
- Presentations and reflective discussions

READING MATERIALS

Recommended Readings

- Graphic Novel By Ahsan Habib

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Class tests and quizzes
- Assignments and creative submissions
- Presentations and mid-term examination

Semester End Examination (40%), including:

- Written and creative assessment evaluating critical understanding, narrative structure, visual composition, and originality

MCT333: CHARACTER DESIGN & COMPOSITION (Z BRUSH)

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT221

TEACHING OBJECTIVES

Character Design & Composition (Z Brush) is designed to introduce students to professional character sculpting workflows using Pixologic ZBrush, focusing on the complete process from base geometry to a finished, textured, and presented character model. The course emphasizes hands-on exploration of sculpting techniques, subdivision workflows, poly modeling, detailing, UV creation, and polypainting while developing an understanding of form, anatomy, stylization, and composition. Students learn how to work efficiently with digital sculpting tools, integrate models from other 3D software, and prepare characters for rendering and presentation. The course prioritizes creative experimentation, industry-relevant workflows, and project-based learning to build confidence in full character production pipelines.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Navigate the Z Brush interface and understand 2.5D and 3D workflows
- Create and manipulate geometry using subdivision levels and sculpting tools
- Apply digital sculpting techniques using brushes, alphas, Dynamesh, and ZSpheres
- Develop stylized character models with detailed forms and surface refinement
- Prepare character models through UV creation, polypainting, rendering, and export

COURSE CONTENT

- Introduction to ZBrush and project overview
- Understanding the ZBrush workflow and 2.5D canvas

- Creating simple base geometry
- Working with files, navigation, and display options
- Subdivision levels, masking, symmetry, and sculpting tools
- Brush systems, strokes, alphas, and sculpting control
- Sculpting workflows using ZSpheres and Dynamesh
- Review and critique session
- Mid-term examination and quizzes
- Stylized character creation workflow
- Reference image setup and base mesh development
- Adding facial features using insert mesh techniques
- High-frequency detailing and surface refinement
- Character posing and layer-based detailing
- Working with materials, lighting, and presentation
- Polypainting using Spotlight and texture painting techniques
- UV creation and output options
- Individual project discussion and problem-solving
- Project presentation, quizzes, and final examination

TEACHING & LEARNING METHODS

- Demonstration-based lectures and guided tutorials
- Hands-on studio sessions in computer labs
- Step-by-step sculpting exercises and workflows
- Project-based learning and character development
- Feedback, critique, and iterative refinement
- Final project presentation and evaluation

READING MATERIALS

Recommended Readings

- Beginner's Guide to Z Brush, 3dtotal Publishing

Supplementary Readings

- Grassetti, Rafael et al. ZBrush Character Sculpting: Volume 1, 3dtotal Publishing

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance and participation
- Quizzes and class tests
- Assignments and sculpting exercises
- Mid-term examination and presentations

Semester End Examination (40%), including:

- Practical and written assessment evaluating sculpting quality, workflow understanding, creativity, and final character presentation

MCT334: SCRIPT PROGRAMMING

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: CSE417

TEACHING OBJECTIVES

This course introduces students to scripting for interactive applications and games using Unity and C#, emphasizing rapid development, object interaction, and real-time behavior control within a game engine. The course focuses on understanding Unity's scripting workflow, core C# syntax as applied to game development, and the logic required to control objects, physics, input, and gameplay mechanics. Through hands-on projects, students learn how scripting languages differ from traditional system programming languages and how they are used to bind game components, automate behaviors, and prototype gameplay features efficiently. By developing a complete mini-game, students gain practical experience in designing, coding, testing, and refining interactive game systems.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply scripting concepts to control game objects and interactions
- Develop gameplay mechanics using C# within the Unity engine
- Design and implement basic 2D and 3D game logic
- Use scripting for real-time interaction, animation, and physics control

COURSE CONTENT

- Introduction to the Unity platform
- Unity engine overview and interface navigation
- Game object creation and scene setup
- Variables, functions, and scripting structure in C#
- Unity lifecycle methods (Awake, Start, Update, FixedUpdate)
- Time-based movement using Delta Time
- User input handling (keyboard, axis, buttons)
- Conditional statements and loops
- Object movement and transformation (Translate, Rotate, LookAt)
- Vector mathematics for motion and direction
- Activating and deactivating GameObjects and components
- Object destruction and point-based interactions

- Instantiation of prefabs and runtime object creation
- Array usage for managing multiple objects
- Bullet mechanics and shooting system development
- Collision handling and object interaction
- Designing reusable scripts and modular components
- Basic class structure and script organization
- Refactoring and improving code quality
- Final project development and evaluation

TEACHING & LEARNING METHODS

- Instructor-led demonstrations and live coding
- Guided lab exercises using Unity and C#
- Hands-on mini-projects and incremental game development
- Problem-solving sessions and debugging practice
- Final game prototype development

READING MATERIALS

Recommended Readings

- Rouse, Richard III. Game Design: Theory and Practice. 2nd Edition

Supplementary Readings

- Harwani, B. M. Learning Object-Oriented Programming in C#
- Dunniway, Troy & Novak, Jeannie. Game Development Essentials: Gameplay Mechanics

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance and participation
- Lab reports and practical exercises
- Lab performance and scripting tasks
- Final game project and demonstration

MCT335: CHARACTER ANIMATION LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40

Prerequisite(s) : MCT313

TEACHING OBJECTIVES

The Character Animation Lab is designed to introduce students to industry-standard character animation workflows using Autodesk Maya, with a primary focus on rigging, motion capture, and animation fundamentals. The course aims to develop practical skills in preparing geometry for animation, creating and applying character rigs using Human IK and Quick Rig systems, and producing believable character motion through walk cycles and motion capture techniques. Through hands-on lab practice, guided demonstrations, and project-based learning, students gain an understanding of professional animation pipelines used in film, television, and game production. The course also emphasizes technical accuracy, creative expression, and efficiency, preparing students to create production-ready animations and enhancing their employability in roles such as Animator, Pre-visualization Artist, Technical Animator, and 3D Generalist.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Rig a biped character using the Human IK system in Autodesk Maya
- Create character rigs using Quick Rig through both one-click and step-by-step methods
- Animate a complete walk cycle demonstrating weight, balance, and timing
- Apply, retarget, and edit motion capture data on Human IK and custom rigs
- Edit animation using the Graph Editor and Time Editor and perform efficient skin weight painting

COURSE CONTENT

- Introduction to character animation workflows and Autodesk Maya interface
- Preparing geometry for rigging and animation
- Overview of rigging concepts and skeletal structure
- Using Quick Rig one-click solution
- Step-by-step Quick Rig setup and skeleton placement
- Binding geometry to skeletons using skinning techniques
- Painting and refining skin weights for hands, legs, and body
- Applying motion capture data to characters
- Editing and refining motion capture animation
- Setting up toon shaders for animated characters
- Using and animating the Human IK rig
- Retargeting raw motion capture data downloaded from external sources
- Walk cycle fundamentals and animation principles
- Walk cycle poses: contact, opposite contact, and passing positions
- Enhancing walk cycles to convey weight and realism
- Introduction to particle goals, colliders, and effect animation
- Creating a complete character animation project
- Final project presentation and evaluation

TEACHING & LEARNING METHODS

- Demonstration-based laboratory sessions
- Hands-on practice using 3D animation tools in computer labs
- Lecture slide presentations and instructional videos
- Brainstorming sessions and open discussions
- Industry pipeline seminars and outdoor learning sessions
- Interactive learning through gamification and digital forums
- Feedback, critique, and question-and-answer sessions

READING MATERIALS

Recommended Readings

- Roberts, Steve. Character Animation in 3D: Use Traditional Drawing Techniques to Produce Stunning CGI Animation. Focal Press.
- Ahmed, Arif. Physics for Animation.

Supplementary Readings

- Murdock, Kelly L. Autodesk 3ds Max 2019 Complete Reference Guide. Pearson Education.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance
- Lab reports and documentation
- Lab performance and practical exercises
- Final practical examination and project submission

MCT336: TEXTURING

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT221

TEACHING OBJECTIVES

The Texturing course is designed to provide students with a strong foundation in 3D texturing workflows for interactive media such as games, animation, and real-time visualization. The

course aims to develop both technical and artistic competencies required to create high-quality textures and materials using industry-standard software including 3ds Max, Maya, and Substance Painter. Through a combination of individual assignments, practical lab sessions, and project-based learning, students learn UV mapping, PBR texturing, material creation, and rendering techniques while following a professional production pipeline. Emphasis is placed on problem-solving, visual consistency, and adaptability in both individual and collaborative environments, enabling students to produce optimized and production-ready assets for contemporary interactive media and game design industries.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Produce simple game-ready 3D assets with appropriate textures following a standard production pipeline
- Create optimized UV layouts and textures compatible with physically based rendering (PBR) and real-time engines
- Demonstrate knowledge of edge flow, mesh optimization, and texturing techniques for hard-surface and organic models
- Apply textures effectively to hard-surface models such as furniture and real-life products
- Create and apply textures on organic models with attention to realism and material behavior

COURSE CONTENT

- Introduction to texturing concepts and workflows in 3D software
- Fundamentals of UVW mapping, UV textures, and unwrapping techniques
- Overview of texturing tools and pipelines
- Creating tile-able textures for game assets
- UV unwrapping techniques in 3ds Max
- UV setup and layout optimization in Maya
- Applying textures to different types of 3D objects
- Introduction to Substance Painter interface and workflow
- Creating materials and custom textures in Substance Painter
- Generating and applying normal maps
- Understanding and implementing PBR materials
- Rendering textured models using Arnold, V-Ray, and Redshift
- Lighting techniques for presenting textured assets
- Applying textures to real-life products and furniture models
- Developing complete texturing projects for hard-surface assets
- Texturing organic models
- Preparing final outputs and presentation-quality renders
- Project documentation and report preparation
- Final project submission and examination

TEACHING & LEARNING METHODS

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

- Interactive lectures with slide presentations and video demonstrations
- Hands-on laboratory sessions using 3D software tools
- Brainstorming sessions and open discussions
- Workshops and guided practical exercises
- Industry-oriented seminars on production pipelines
- Outdoor and experiential learning sessions
- Gamified learning activities and interactive online forums
- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Ahmed, Arif. Physics for Animation.

Supplementary Readings

- Murdock, Kelly L. Autodesk 3ds Max 2019 Complete Reference Guide. Pearson Education.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance
- Lab reports and documentation
- Lab performance and practical exercises
- Final project submission and practical examination

CSE414: SIMULATION AND MODELING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT451

TEACHING OBJECTIVES

The Simulation and Modeling course is designed to introduce students to the principles, theory, and practical application of fluid dynamics and simulation for visual effects using industry-standard software. The course aims to develop a strong conceptual understanding of physical properties such as density, viscosity, pressure, surface tension, and scale while enabling students to apply these concepts through hands-on simulations. Emphasis is placed on working

with particle systems, emitters, forces, and large-scale fluid solvers to create realistic effects ranging from small splashes to massive ocean simulations. Through guided demonstrations, practical exercises, and project-based learning, students gain exposure to professional simulation pipelines used in films, television, commercials, and visual effects production, preparing them to create technically accurate and visually compelling fluid-based animations.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Work efficiently with RealFlow as a professional fluid simulation tool
- Apply fundamental fluid simulation theories to practical visual effects scenarios
- Create small-scale fluid simulations with varying viscosity and physical parameters
- Develop large-scale simulations involving high particle counts and complex dynamics
- Produce fluid-based effect animations suitable for visual effects and media production

COURSE CONTENT

- Introduction to simulation and modeling concepts
- Overview of RealFlow interface, viewport navigation, and project setup
- Understanding fluid and object dynamics
- Types of emitters, daemons, and force systems
- Particle systems, emitters, and daemons: fundamentals
- Circle emitter setup and particle parameter control
- Advanced emitter types: square, sphere, linear, triangle, cylinder, and object emitters
- Importing and exporting mesh objects between 3D software and RealFlow
- Scaling, adjusting meshes, and simulation setup
- Using gravity, collision, K-volume, K-age, and K-speed daemons
- Creating controlled fluid simulations using force combinations
- RealWave system and water surface generation
- Controlling wave parameters such as polygon size, damping, and depth
- Mid-term assessment and quizzes
- Particle resolution optimization and simulation quality control
- Controlling density, viscosity, pressure, surface tension, and interpolation
- Managing particle velocity for animated fluid flow
- Converting particles to mesh and mesh optimization techniques
- Controlling mesh properties such as surface proximity, thinning, relaxation, and tension
- Simulation of waterfalls, rivers, and flowing water effects
- Large-scale fluid simulations using Hybrid technology
- Open and closed domain simulations
- Creating splash, foam, and secondary particles
- Exporting simulations using Alembic for rendering in 3D software
- Developing a complete fluid simulation project
- Final assessment and examination

TEACHING & LEARNING METHODS

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

- Interactive lectures with slide presentations
- Demonstration-based laboratory sessions
- Hands-on simulation exercises using RealFlow
- Open discussions and problem-solving sessions
- Guided experimentation with particle systems and forces
- Continuous feedback and critique sessions
- Project-based learning and practical assessments

READING MATERIALS

Recommended Readings

- Keller, Eric. Maya Visual Effects: The Innovator's Guide. Autodesk Official Press.
- Ahmed, Arif. Physics for Animation.

Supplementary Readings

- Fisher, Gordon. Blender 3D Basics, 2nd Edition.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating conceptual understanding, application, analysis, and creative execution

CSE415: SIMULATION AND MODELING LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT335

TEACHING OBJECTIVES

The Simulation and Modeling Lab is designed to provide hands-on training in advanced fluid simulation techniques using RealFlow, with a strong focus on Hybrid-based large-scale water

simulations and secondary particle systems. The course aims to develop practical expertise in creating complex dynamic effects such as stormy oceans, splashes, foam, mist, and tsunami-like interactions while integrating rigid body dynamics and controller-driven animation. Through guided laboratory exercises and project-based learning, students gain experience in importing and exporting simulation data, interacting objects with fluids, and managing high-resolution simulations suitable for professional visual effects pipelines. The course emphasizes realism, scale, technical control, and industry-relevant workflows, preparing students to create production-ready fluid simulations for film, television, and digital media.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Understand and apply the principles of Hybrid fluid simulation in RealFlow
- Create and control large-scale fluid simulations with splash and foam systems
- Export and integrate simulated data into 3D software for final output
- Simulate object–fluid interactions including tsunami and impact-based effects
- Produce a complete visual effects animation project using fluid dynamics

COURSE CONTENT

- Introduction to Hybrid simulation principles
- Understanding hybrid fluid systems in RealFlow
- Creating Hybrid-based water simulations
- Converting Hybrid fluids into mesh objects
- Controlling mesh parameters for large-scale simulations
- Importing 3D scenes from external software into RealFlow
- Simulating large-scale fluids using Hybrid, splash, and foam systems
- Managing secondary particle effects such as splashes and foam
- Exporting simulated mesh and particle data to 3D software
- Preparing simulation outputs for rendering
- Interacting rigid objects with hybrid fluid systems
- Creating tsunami and large-impact water effects
- Exporting simulations using Alembic workflow
- Developing a complete tsunami-style fluid simulation project
- Refining and optimizing simulations for final presentation
- Final project submission and examination

TEACHING & LEARNING METHODS

- Demonstration-based laboratory sessions
- Hands-on practice using RealFlow simulation tools
- Lecture slide presentations and instructional videos
- Brainstorming sessions and open discussions
- Workshops and guided practical exercises
- Industry-oriented seminars and pipeline discussions
- Gamified learning activities and interactive online forums

- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Ahmed, Arif. Physics for Animation.

Supplementary Readings

- Murdock, Kelly L. Autodesk 3ds Max 2019 Complete Reference Guide. Pearson Education.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance
- Lab reports and documentation
- Lab performance and practical execution
- Final project submission and practical examination

MCT413: NEW TYPEFACE DESIGN

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT133

TEACHING OBJECTIVES

The New Typeface Design course is designed to develop advanced conceptual, analytical, and practical skills in alphabet design and typography, with a focus on creating original typefaces in both Bengali and English scripts. The course aims to deepen students' understanding of typographic anatomy, historical evolution, cultural context, and digital transformation of letterforms while positioning typography as a research-driven design discipline. Through structured exploration, critical analysis, and studio-based practice, students learn to conceptualize, design, refine, and publish complete Unicode-compatible font systems. Emphasis is placed on originality, precision, craftsmanship, and research-informed decision-making, enabling students to transition from conventional typographic practice to sophisticated typeface design suitable for professional and national-level contributions.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Demonstrate comprehensive knowledge of typographic terminology and principles
- Analyze and apply the anatomical structures of Bengali and English alphabets
- Design complete alphabet systems based on sound typographic anatomy
- Develop and complete a Unicode-compatible font as a final project
- Understand and apply the full workflow of professional font development and publication

COURSE CONTENT

- Introduction to new typeface design and contemporary typography practice
- Historical and cultural evolution of alphabets
- Typography as a medium of expression and research
- Experience sharing, motivation, and industry overview
- Review of typographic terminology and foundational principles
- Elements and anatomy of typography
- Relationship between prerequisite typography courses and advanced practice
- Concept development and pre-learning in font design
- Idea generation and innovation in alphabet creation
- Free-form character sketching and exploratory drawing
- Partial font development and capability assessment
- Mid-term examination and quizzes
- Group-based font selection and critical review
- Comparative analysis of individual and group designs
- Collaborative decision-making for complete font development
- Project-based studio interaction and guided refinement
- Error identification and typographic problem-solving
- Continuous feedback and iterative improvement
- Font lab studio introduction and workflow demonstration
- Methods of font construction and Unicode implementation
- Font publishing processes and best practices
- Final project submission, examination, and evaluation

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and videos
- Brainstorming and guided discussion sessions
- Studio-based typography workshops
- Use of 2D graphic and font development tools
- Outdoor and experiential learning sessions
- Industry-oriented seminars on typography and design practice
- Project-based learning and peer critique
- Gamification and interactive digital forums
- Continuous feedback, mentoring, and critique sessions

READING MATERIALS

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

Recommended Readings

- The Complete Manual of Typography, Second Edition. Peachpit Press.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating conceptual understanding, analysis, application, and creative output

MCT411: DIGITAL MATTE PAINTING

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT233

TEACHING OBJECTIVES

The Digital Matte Painting course is designed to introduce students to the principles and practical workflows of matte painting and compositing for film, television, and digital media. The course aims to develop foundational skills in photo and video manipulation, chroma keying, rotoscoping, lighting, shading, and compositing using industry-standard tools such as Adobe Photoshop and Adobe After Effects. Through hands-on laboratory sessions and industry-referenced examples, students learn to integrate 2D and 3D elements into visually convincing environments while understanding the technical and artistic considerations of cinematic composition. Emphasis is placed on realism, visual continuity, and professional production practices, enabling students to create polished matte paintings and composited shots suitable for various media outputs.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Use the interfaces of Adobe Photoshop and Adobe After Effects effectively for matte painting

- Apply photo and video manipulation techniques for visual effects production
- Manage layers efficiently in both 2D and 3D compositing environments
- Apply appropriate lighting and shading techniques to enhance realism
- Composite rendered elements and footage to produce complete matte painting outputs

COURSE CONTENT

- Introduction to digital matte painting concepts and workflows
- History and industrial applications of matte painting in cinema
- Professional and Hollywood matte painting case studies
- Adobe Photoshop interface and tools for matte painting
- Adobe After Effects interface and compositing workflow
- Understanding cinematic composition and visual storytelling
- Layer management techniques for matte painting
- Working with solid, adjustment, null, and shape layers
- Chroma keying techniques using green and blue screen footage
- Background removal and clean plate creation
- Rotoscoping techniques for object and character isolation
- Lighting and shading principles for composited environments
- Color correction and visual consistency
- Integrating 3D elements into 2D matte paintings
- Compositing workflows combining rendered 3D assets and live-action footage
- Output preparation for different media formats
- Quizzes, final practical examination, and project submission

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and video demonstrations
- Hands-on laboratory sessions using matte painting software tools
- Analysis of professional and cinematic examples
- Outdoor sessions involving chroma, camera, and lighting setups
- Guided practical exercises and demonstrations
- Gamified learning activities and interactive engagement
- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Mattingly, David B. The Digital Matte Painting Handbook. Sybex.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance
- Lab reports and documentation
- Lab performance and practical exercises

- Final practical examination and project submission
-

MCT421: GAME ANALYSIS AND DEVELOPMENT

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT431

TEACHING OBJECTIVES

The Game Analysis & Development course is designed to equip students with conceptual, analytical, and practical knowledge of modern game design, development, and commercialization across mobile and computer platforms. The course aims to develop creative and critical thinking skills by examining user behavior, cognitive psychology, interaction design, game mechanics, artificial intelligence, and monetization strategies. Through a combination of theoretical exploration, case studies, and hands-on development using contemporary game engines, students learn to analyze successful and unsuccessful games, design engaging gameplay experiences, and select appropriate platforms and technologies. Emphasis is placed on user-centered design, market awareness, and technical feasibility, enabling students to conceptualize, develop, and promote interactive games that align with industry standards and player expectations.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Develop a conceptual framework for designing multiplatform games
- Analyze problem statements to construct effective game architectures and mental models
- Apply artificial intelligence concepts within game environments
- Research, plan, and create promotional and monetization strategies for games

COURSE CONTENT

- Introduction to game analysis and development
- Overview of the game industry and interactive entertainment
- Modeling gamer profiles and understanding player behavior
- Human psychophysiology, perception, and cognitive biases
- Motivation, goal-setting, gamification, progress, and achievements
- Symbolic systems, applied semiotics, localization, and internationalization
- User-oriented and player-centric design principles

- Overview of gaming platforms and development environments
- Introduction to Unity game engine
- Game classification and lifecycle
- Technology stacks and major platform ecosystems (Microsoft, Google, Apple)
- Overview of game development frameworks and engines
- Fundamentals of game mechanics and interaction design
- Artificial intelligence concepts for single-player and multiplayer games
- Game UX principles and usability considerations
- Game distribution platforms and marketplaces
- Monetization strategies and revenue models
- Mobile advertising technologies and analytics tools
- Game feedback systems, rankings, and KPI analysis
- Branding, franchising, and marketing best practices
- Principles of game level design
- Designing cooperative and competitive gameplay experiences
- Creating rooms, instances, and interactive levels
- Introduction to AI systems and waypoint navigation
- Vehicle AI systems and behavior customization
- Mid-term examination and quizzes
- Character development and avatar design
- Player–avatar relationships and narrative roles
- Movement systems, audio integration, animation, and AI
- Case study: Art gallery game development
- Functional workflows, diagrams, and project implementation
- UI and UX design for games
- Interaction models, camera systems, and navigation mechanisms
- Augmented Reality (AR) and Virtual Reality (VR) in games
- Final examination and assessment

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and video materials
- Brainstorming and open discussion sessions
- Case studies and academic debates
- Hands-on practical sessions and guided development
- Mind-mapping and conceptual modeling exercises
- Gamified learning activities and interactive platforms
- Continuous feedback, mentoring, and Q&A sessions

READING MATERIALS

Recommended Readings

- Brockschmidt, Kraig. Programming Windows Store Apps with HTML, CSS, and JavaScript. Microsoft Press, 2014.

- Genadinik, Ales. Mobile App Marketing and Monetization. Semantic Valley LLC, 2014.
- UBM. Gamasutra: The Art & Business of Making Games.

Supplementary Readings

- Petzold, Charles. Creating Mobile Apps with Xamarin.Forms. Microsoft Press, 2015.
- Schell, Jesse. The Art of Game Design: A Deck of Lenses. Schell Games, 2008.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating conceptual understanding, application, analysis, and creative problem-solving

MCT422: GAME ANALYSIS AND DEVELOPMENT LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT431

TEACHING OBJECTIVES

The Game Analysis & Development Lab is designed to provide hands-on technical training in 2D game development with a strong focus on programming, problem-solving, and industry-standard workflows. The course aims to develop students' competencies in game programming using C#, JavaScript, and HTML5 while introducing mathematical foundations such as vectors and matrices essential for real-time applications. Through structured laboratory exercises and guided projects, students learn core aspects of game architecture, user input handling, graphics, audio, collision detection, and publishing pipelines. The course also emphasizes software development lifecycles, teamwork, and best practices for memory- and performance-constrained platforms, enabling students to independently design and develop complete 2D arcade-style games and prepare for diverse career paths within the game industry.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Demonstrate intermediate-level programming and problem-solving skills for game development
- Develop 2D games using C#, JavaScript, and HTML5
- Apply mathematical concepts such as vectors and matrices in real-time game simulations
- Design and develop complete 2D arcade-style games following industry workflows

COURSE CONTENT

- Introduction to game development lab workflow and tools
- JavaScript and HTML5 fundamentals for games
- Project design, organization, and coding standards
- Programming language fundamentals and data types
- Object-oriented programming concepts and review
- Containers, exceptions, delegates, and event handling
- String manipulation and memory considerations
- Mathematical foundations for games
- Cartesian coordinate systems and spatial representation
- World space, object space, and camera space
- Vector and matrix operations for movement and transformation
- Fundamentals of game design theory
- Cultural and functional definitions of games
- Game development processes and documentation practices
- Technical aspects of game development
- User input handling and control systems
- Graphics and rendering basics
- Audio integration in games
- Game engine architecture
- Collision detection techniques
- Publishing workflows and indie development practices
- Introduction to content pipelines
- User interface development for games
- Unity UI elements including buttons and controls
- UI design principles for player interaction
- Game art fundamentals
- Perspective, color theory, and image editing
- Publishing games for PC, Mac, Linux, smartphones, and consoles
- Splash screens, credit scenes, and icons
- Software development lifecycles
- Team collaboration, documentation, and development tools
- Final project development and examination

TEACHING & LEARNING METHODS

- Demonstration-based laboratory sessions
- Hands-on programming and game development exercises
- Interactive lectures with slide presentations and video tutorials
- Brainstorming sessions and guided discussions
- Case studies and academic debates
- Mind-mapping and design planning activities
- Gamified learning and interactive digital tools
- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Halpern, Jared. Developing 2D Games with Unity: Independent Game Programming with C#. 2018.

Supplementary Readings

- Jackson, Simon. Mastering Unity 2D Game Development, Second Edition.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance
 - Lab reports and documentation
 - Lab performance and practical execution
 - Final project submission and practical examination
-

MCT424: AUGMENTED AND VIRTUAL REALITY (AR/VR) DEVELOPMENT

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT421, MCT422

TEACHING OBJECTIVES

The Augmented and Virtual Reality (AR/VR) Development course is designed to provide students with a comprehensive understanding of immersive technologies, their historical evolution, technical foundations, and experiential design principles. The course aims to

cultivate analytical and practical skills required to design, develop, and evaluate AR, VR, and mixed reality applications across diverse domains such as entertainment, education, healthcare, industrial visualization, and research. Through theoretical exploration, case study analysis, and hands-on development using contemporary AR/VR tools and platforms, students learn to create immersive experiences that integrate interaction design, real-time graphics, tracking technologies, and user-centered design principles. Emphasis is placed on collaborative development, prototyping, ethical considerations, and emerging industry practices, preparing students to contribute effectively to this rapidly evolving and commercially viable field.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Demonstrate a comprehensive understanding of AR and VR design and development principles
- Analyze user experience and interaction design in immersive AR/VR environments
- Design and create interactive AR and VR applications using contemporary frameworks and tool
- Conduct research and critical analysis related to AR/VR technologies, platforms, and applications

COURSE CONTENT

- Overview of augmented, virtual, and mixed reality
- Fundamentals of AR/VR systems and immersive media
- Real-time examples and industry analysis
- Introduction to augmented reality technologies and applications
- AR use cases across industries
- Tools and software for AR development on mobile platforms
- Development environment setup for AR applications
- Vuforia configuration and integration with Unity
- Challenges and limitations in AR development
- Menu design principles for immersive applications
- Image-based menus and interactive UI elements
- Animated menu systems for AR/VR applications
- Lighting principles for immersive environments
- Rendering techniques for real-time AR/VR experiences
- Introduction to AR toolkits and SDKs
- Overview and setup of ARKit, Vuforia, EasyAR SDK, and ARCore
- Review and consolidation of AR concepts
- Mid-term examination and quizzes
- Image tracking, multi-image tracking, and cube tracking techniques
- Developing AR applications in Unity 3D
- Creating and deploying AR mobile applications
- Integrating user interactions into AR applications
- ARKit integration and tracking workflows

- Introduction to virtual reality technologies
- VR applications across different industries
- Tools and software for VR development
- Setting up VR development environments
- Challenges and constraints in VR design and development
- Game market strategy and monetization analysis
- Revenue models and publishing strategies for immersive applications
- Multi-platform deployment and analytics
- Final examination and assessment

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and video demonstrations
- Brainstorming sessions and open discussions
- Hands-on practical sessions using AR/VR tools and headsets
- Case studies and industry-focused seminars
- Collaborative group projects and prototype development
- Workshops and guided laboratory exercises
- Gamified learning activities and interactive online platforms
- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Hale, Kelly S., and Stanney, Kay M. (Eds.). Handbook of Virtual Environments: Design, Implementation, and Applications, Second Edition.

Supplementary Readings

- Madary, Michael, and Metzinger, Thomas K. Real Virtuality: A Code of Ethical Conduct. Frontiers in Robotics and AI, 2016.
- Jerald, Jason. The VR Book: Human-Centered Design for Virtual Reality. ACM and Morgan & Claypool, 2015.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination (25%)

Semester End Examination (40%), including:

- Written and practical assessment evaluating understanding, application, analysis, and creative development of AR/VR solutions

MCT431: MULTIMEDIA AUTHORIZING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT334

TEACHING OBJECTIVES

The Multimedia Authoring course is designed to develop advanced skills in designing, programming, and managing interactive multimedia applications with a strong focus on game development and interactive systems. The course aims to equip students with conceptual, technical, and creative competencies in multimedia programming, object-oriented authoring, scripting techniques, and the integration of diverse media types such as text, graphics, animation, audio, and video. Through a combination of theoretical discussion, practical implementation, and project-based learning, students gain experience in interface design, game mechanics, level design, and cross-platform development strategies. Emphasis is placed on problem-solving, efficient programming practices, production workflows, and the creative potential of multimedia authoring, enabling students to conceive, design, and implement complex interactive applications suitable for modern digital platforms.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Demonstrate an understanding of game design principles and multimedia programming concepts
- Analyze problem statements to develop structured game architectures and design models
- Apply object-oriented programming techniques to develop interactive multimedia applications and games
- Conduct research and critical analysis related to game design, development, and multimedia authoring

COURSE CONTENT

- Introduction to multimedia authoring and interactive media
- Overview of the gaming sector in Bangladesh and global markets
- Evolution of the worldwide gaming industry and revenue models
- Developing games for modern platforms
- Basic of Game UX/UI: Menu design fundamentals using images using buttons, colors, and animations.
- Lighting concepts for interactive media

- Rendering techniques in game development
- First-person character development
- Understanding player behavior and interaction
- Camera setup and control systems
- Character movement and input handling
- Programming character behavior and mouse control
- Mid-term examination and quizzes
- Animation principles and workflows in Unity
- Animation blending and mixing techniques
- Scripting and controlling animations programmatically
- Shooting game development fundamentals
- Player movement and navigation systems
- Weapon setup, ray casting, and shooting mechanics
- Visual and audio feedback systems
- Ammunition management, scoring, and UI display
- Coin systems, restart mechanics, and game logic
- Weapon selling and progression systems
- Level design principles and processes
- Layout planning and iterative level development
- 2D game development concepts
- Designing and programming 2D games
- Case study: Flappy Bird-style game development
- Scoring, collision, and object destruction
- Game market strategy and monetization analysis
- Revenue models and publishing strategies
- Multi-platform publishing and analytics
- Final examination and assessment

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and video demonstrations
- Brainstorming sessions and open discussions
- Hands-on practical sessions and guided programming exercises
- Case studies and academic debates
- Mind-mapping and conceptual modeling activities
- Gamified learning approaches and interactive tools
- Continuous feedback, mentoring, and Q&A sessions

READING MATERIALS

Recommended Readings

- Brockschmidt, Kraig. Programming Windows Store Apps with HTML, CSS, and JavaScript. Microsoft Press, 2014.

- Genadinik, Ales. Mobile App Marketing and Monetization. Semantic Valley LLC, 2014.
- UBM. Gamasutra: The Art & Business of Making Games.

Supplementary Readings

- Petzold, Charles. Creating Mobile Apps with Xamarin.Forms. Microsoft Press, 2015.
- Schell, Jesse. The Art of Game Design: A Deck of Lenses. Schell Games, 2008.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating conceptual understanding, application, analysis, and creative development

MCT432: DIGITAL CINEMATOGRAPHY

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT233

TEACHING OBJECTIVES

The Digital Cinematography course is designed to develop students' technical, aesthetic, and analytical understanding of modern electronic cinematography through intensive practical and conceptual exploration. The course aims to establish a strong foundation in shooting techniques, digital camera systems, lighting design, shading methods, and visual composition while emphasizing storytelling through moving images. Students learn to integrate camera operation, lens choice, lighting grammar, color, and post-production workflows to create visually compelling cinematic outputs. Through hands-on exercises, critical discussions, and project-based learning, students develop the ability to analyze, critique, and apply digital cinematography principles across creative and research-driven contexts, while documenting their process and artistic growth through structured multimedia presentations.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Apply professional shooting techniques for digital cinematography
- Create multimedia presentations integrating cinematic visuals
- Use rendering and export techniques for high-quality digital output
- Apply lighting concepts and shading techniques to enhance visual storytelling
- Demonstrate a comprehensive understanding of digital cinematography workflows

COURSE CONTENT

- Introduction to 35mm film cinematography and digital cinematography
- Comparison between analog and digital cinematic workflows
- Video formats: SD, HD, 2K, UHD, 4K, and emerging standards
- Digital sensors: CCD and CMOS fundamentals
- Signal processing, color sampling, sampling rate, and pixel resolution
- Understanding bits, bytes, and data depth
- Recording methods: scanning, interlace, and progressive
- Storage and recording media: tape, card, FMU
- Compressed and uncompressed recording techniques
- Camera user menu and operational controls
- Filters, white balance, black balance, and gain (dB)
- ISO rating, frame rate, shutter speed, and shutter angle
- Cine gamma, exposure, and exposure latitude
- Mid-term examination and quizzes
- Lens theory: zoom and prime lenses
- Focus systems, focal length, angle of view, and depth of field
- Hyperfocal distance and lens filters
- Lighting tools and techniques
- Three-point lighting and grammar of light
- Light meters: incident and reflected readings
- Camera support systems: tripod, dolly, jib, crane, steadicam, and drone
- Classification of shots and the 5 C's of cinematography
- Aesthetic principles and golden rules of composition
- Advanced composition understanding
- Blending modes and track matte techniques
- Compositing real footage
- Exporting and rendering using After Effects and Media Encoder
- Final presentations, quizzes, and examination

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and video demonstrations
- Hands-on practical shooting sessions
- Outdoor field exercises with camera and lighting equipment
- Laboratory sessions using video editing and compositing software

- Guided demonstrations and technical workshops
- Peer critique, feedback, and reflective discussions
- Gamified learning activities and formative assessments

READING MATERIALS

Recommended Readings

- Stump, David. Digital Cinematography: Fundamentals, Tools, Techniques, and Workflows. 2014.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating conceptual understanding, technical execution, analysis, and creative application

MCT433: LIGHTING AND RENDERING LAB

Course Type	: Lab
Credits	: 1
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT321

TEACHING OBJECTIVES

The Lighting and Rendering Lab is designed to develop students' practical and conceptual skills in digital lighting, shading, and rendering using the V-Ray rendering engine within 3ds Max. The course aims to familiarize students with professional lighting workflows for interior and exterior environments while emphasizing photorealism, visual storytelling, and technical accuracy. Through hands-on, project-based exercises, students learn to create realistic daylight and nighttime scenes, apply physical camera settings, balance artificial and environmental lights, and enhance rendered outputs using compositing techniques. The course also focuses on strengthening students' understanding of lighting theory, material behavior, and rendering

principles, enabling them to efficiently utilize advanced rendering tools and produce customized lighting and shading solutions suitable for architectural visualization, animation, and game environments.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Create advanced V-Ray materials and realistic textures
- Design and implement effective lighting setups for interior and exterior scenes
- Use the V-Ray Physical Camera to achieve accurate exposure and composition
- Apply professional-level rendering workflows using the V-Ray render engine
- Produce photorealistic still images and animated visualizations

COURSE CONTENT

- Introduction to lighting and rendering concepts in V-Ray
- Overview of V-Ray interface and render settings
- Setting up exterior daylight scenes
- Sunlight and environmental lighting techniques
- Creating photorealistic exterior renders
- Managing scene elements such as vegetation and water surfaces
- Developing full village or city-scale environments
- Daylight setup with exposure compensation
- Use of auxiliary and fill lights
- Balancing multiple light sources
- Rendering large-scale exterior scenes
- Setting up nighttime lighting scenarios
- Color correction and lens effects for night renders
- Nighttime lighting aesthetics and mood creation
- Rendering and compositing night scenes
- Architectural visualization project planning
- Applying V-Ray lighting, materials, and camera systems
- Rendering interior and exterior architectural scenes
- Creating animated walkthroughs (minimum 60 seconds, HD format)
- Final project refinement and presentation
- Final practical examination

TEACHING & LEARNING METHODS

- Demonstration-based laboratory sessions
- Hands-on practice using V-Ray and 3ds Max
- Lecture slide presentations and instructional videos
- Brainstorming sessions and open discussions
- Workshops and guided practical exercises
- Industry-oriented seminars on rendering pipelines
- Gamified learning activities and interactive online forums

- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Keller, Eric. Maya Visual Effects: The Innovator's Guide. Autodesk Official Press.
- Ahmed, Arif. Physics for Animation.

Supplementary Readings

- Boughen, Nicholas. 3ds Max Lighting.

EVALUATION METHODS

Continuous Internal Evaluation (100%), including:

- Attendance
- Lab reports and documentation
- Lab performance and practical execution
- Final project submission and practical examination

MCT450: LIGHTING AND RENDERING

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT321

TEACHING OBJECTIVES

The Lighting and Rendering course is designed to develop students' theoretical understanding and practical mastery of digital lighting, shading, and rendering using the V-Ray rendering engine in 3ds Max. The course aims to explore the physical, mathematical, and aesthetic principles that govern the interaction of light, texture, and form in computer graphics, with a strong emphasis on visual storytelling and cinematic realism. Through a combination of conceptual lectures, hands-on exercises, and project-based learning, students learn to design and implement professional lighting setups for interior and exterior environments, create advanced materials, and optimize rendering workflows. The course encourages experimentation across a range of visual styles—from photorealistic to stylized—while equipping students with the skills required for architectural visualization, animation, and game production.

EXPECTED LEARNING OUTCOMES

Daffodil International University
Daffodil Smart City, Birulia, Savar, Dhaka-1216
Web: www.daffodilvarsity.edu.bd

Upon successful completion of the course, students will be able to:

- Create advanced-level materials and realistic textures using V-Ray
- Design and implement effective lighting setups for diverse scenes
- Configure and use physical camera settings for accurate exposure and composition
- Apply professional-level rendering workflows using the V-Ray render engine
- Explore and utilize advanced V-Ray features for optimized and high-quality output

COURSE CONTENT

- Introduction to lighting and rendering concepts
- V-Ray basic setup and render engine overview
- Primary and secondary bounce engine configuration
- V-Ray Frame Buffer setup for interactive rendering
- Creating a basic scene for lighting studies
- Setting up portal lights using V-Ray Plane Lights
- Exploring V-Ray Plane Light parameters: multiplier, shadows, color, and temperature
- Inverse-square law of light and its application
- Three-point lighting technique: key, fill, and rim/back light
- Creating and balancing multiple light sources
- Using IES (Illuminating Engineering Society) lights in interior scenes
- Dome light setup and HDRI-based lighting for interiors and exteriors
- Sphere lights and combining multiple light types
- Render optimization and reducing render time
- Review and consolidation of lighting concepts
- Mid-term examination
- Creating V-Ray materials
- Glass, metal, water, and custom material development
- Material refinement through lighting and render parameter combinations
- Setting up interior daylight scenes
- Lighting and rendering furnished interior environments
- Creating photorealistic interior outputs
- Architectural visualization project planning
- Advanced use of V-Ray lighting, materials, and rendering
- Producing a minimum 60-second HD animated walkthrough
- Final project refinement and review
- Final examination

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and video demonstrations
- Hands-on practical sessions using V-Ray and 3ds Max
- Brainstorming sessions and open discussions
- Workshops and guided technical exercises
- Industry-oriented seminars on rendering pipelines
- Gamified learning activities and interactive forums

- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Keller, Eric. Maya Visual Effects: The Innovator's Guide. Autodesk Official Press.
- Ahmed, Arif. Physics for Animation.

Supplementary Readings

- Boughen, Nicholas. 3ds Max Lighting.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating conceptual understanding, technical application, analysis, and creative execution

MCT451: CHARACTER ANIMATION

Course Type	: Theory
Credits	: 3
Duration	: One semester
Total Contact Hours	: 40
Prerequisite(s)	: MCT313

TEACHING OBJECTIVES

The Character Animation course is designed to develop students' conceptual understanding and practical expertise in 3D character animation for games, animated films, and applied visualization domains such as healthcare and life sciences. The course aims to guide students through the complete character animation pipeline, including character modeling, rigging, skinning, and animation using industry-standard tools. Emphasis is placed on planning animation based on scripts and storyboards, constructing efficient rigging systems with FK and IK controls, applying deformation techniques for realistic motion, and animating characters through walk cycles, actions, and expressive performances. Through hands-on exercises,

project-based learning, and critical feedback, students gain the ability to animate characters convincingly and integrate them into complete animated sequences.

EXPECTED LEARNING OUTCOMES

Upon successful completion of the course, students will be able to:

- Create and animate 3D characters using complete control systems
- Animate procedural walk cycles, actions, poses, and expressive movements
- Use character studio tools effectively for animation production
- Model 3D characters suitable for animation
- Perform rigging and animation workflows for complete 3D character animation projects

COURSE CONTENT

- Introduction to character animation and production pipeline
- Character modeling fundamentals in 3ds Max
- Overview of Character Studio and Biped system
- Creating and modifying biped structures
- Walk cycle fundamentals and basic animation principles
- Posing bipeds and setting biped parameters
- Fitting biped structures to character meshes
- Working with non-human bipeds
- Skinning techniques using Physique and Skin modifiers
- Associating meshes with skeletons and test animations
- Correcting deformation and weighting issues
- Footstep animation fundamentals
- Creating and editing footsteps using the dope sheet
- Timing, placement, and dynamics of footstep animation
- Animating movement along circular paths
- Understanding centripetal and centrifugal forces in animation
- Using bank parameters for natural motion
- Differences between auto key and set key animation
- Copying and pasting animation controllers
- Freeform animation techniques
- Forward and inverse kinematics (FK/IK) principles
- Animating IK settings and blending FK/IK systems
- Mid-term examination and quizzes
- Body animation using keyframes
- Working with animation layers and incremental adjustments
- Copying and modifying postures
- Animating arms and hands
- Linking objects to bipeds and pinning hands to props
- Creating follow and constraint-based animations
- Combining multiple motions

- Motion Mixer and Motion Flow techniques
- Saving and managing animation states
- Introduction to character rigging in Maya
- Integrating animation into final output
- Final project development and examination

TEACHING & LEARNING METHODS

- Interactive lectures with slide presentations and video demonstrations
- Hands-on laboratory sessions using 3D animation tools
- Brainstorming sessions and open discussions
- Workshops and guided animation exercises
- Industry-oriented seminars on animation pipelines
- Outdoor and experiential learning sessions
- Gamified learning activities and interactive forums
- Continuous feedback, critique, and Q&A sessions

READING MATERIALS

Recommended Readings

- Roberts, Steve. Character Animation in 3D: Use Traditional Drawing Techniques to Produce Stunning CGI Animation. Focal Press.
- Ahmed, Arif. Physics for Animation.

Supplementary Readings

- Giesen, Rolf, and Khan, Anna. Acting and Character Animation: The Art of Animated Films, Acting, and Visualizing.

EVALUATION METHODS

Continuous Internal Evaluation (60%), including:

- Attendance
- Class tests and quizzes
- Assignments
- Presentations
- Mid-term examination

Semester End Examination (40%), including:

- Written and practical assessment evaluating conceptual understanding, animation principles, technical execution, and creative performance

MCT499: FINAL PROJECT

Course Type : Project

Credits	: 6
Duration	: Two semesters
Total Contact Hours	: 80
Prerequisite(s)	: MIS575

DESCRIPTION

The Final Project is the penultimate academic requirement of the Bachelor of Science degree and represents the culmination of students' learning across the entire program. Spanning two consecutive semesters and carrying six credits, this course provides students with the opportunity to independently conceptualize, design, develop, and present a substantial project that demonstrates their technical competence, creative ability, research aptitude, and professional readiness. Students are expected to identify a meaningful problem or opportunity within their field of study, conduct relevant background research, and propose an appropriate solution grounded in theoretical knowledge and practical skills acquired throughout the program.

The project is conducted in two structured phases. In the first phase, students focus on problem definition, literature review, requirement analysis, project planning, and initial design and development. This phase culminates in a Pre-Defense, where students formally present their interim progress, methodology, and preliminary outcomes before an academic panel. Feedback received during the pre-defense is used to refine the project's technical, conceptual, and aesthetic aspects. The second phase emphasizes full implementation, testing, validation, documentation, and final refinement of the project, leading to the Final Defense, where students demonstrate the completed work and defend their design decisions, implementation strategies, and results.

Throughout the course, students work under the guidance of a faculty supervisor and are expected to demonstrate independent learning, critical thinking, problem-solving, time management, and ethical responsibility. Equal importance is placed on both the process and the final outcome, including project documentation, presentation quality, and the ability to articulate ideas clearly in written and oral form. The Final Project prepares students for professional practice, entrepreneurship, and postgraduate studies by simulating real-world development and research environments and by fostering confidence, accountability, and interdisciplinary integration.